Decarbonizing Sette Laghi
Technical, Economic, and Environmental Study

Politecnico di Milano

Alessandro Borgonovo, Gaia Larici, Riccardo Monti, Simone Pagliuca, Riccardo Ronchi
borgo, gaia, riccardo, 10698534, richard

Course: Integration of Nuclear and Renewables for Carbon Neutral Scenarios
Academic Year: 2024/2025

ABSTRACT

This study evaluates the technical and economic feasibility of integrating Small Modular Reactors (SMRs) into the residential energy sector of the north-western province of Varese, Italy. The proposed hybrid system combines nuclear power with renewable energy sources, hydrogen production, and synthetic methane generation to meet regional energy demands and support decarbonization efforts. The analysis includes dynamic simulations of the plant's operation, economic assessments, and environmental impact evaluations over a 60-year period. The methodology involves data collection from regional energy databases, dynamic modeling using Dymola software, and economic analysis considering capital and operational expenditures, as well as revenue streams from electricity, hydrogen, and synthetic gas sales. Preliminary results indicate the potential for significant decarbonization and energy stability, although further analysis is required to determine the economic viability and optimal configuration of the system. [Placeholder for conclusions: The study concludes with insights into the overall feasibility and potential improvements for the proposed hybrid energy system.]

Key-words: Small Modular Reactors (SMRs), Hybrid Energy Systems, Decarbonization, Hydrogen Production, Synthetic Methane, Dynamic Simulation, Economic Feasibility, Environmental Impact, Residential Energy Sector, Renewable Energy Integration

CONTENTS

1 INTRODUCTION

1.1 Context

Italy is undergoing a substantial energy transition aimed at achieving net-zero emissions by 2050. The residential sector is pivotal in this transition, accounting for approximately $\sim 22\%$ of the total energy consumption and emissions [1].

The residential sector exhibits significant potential for decarbonization through the adoption of renewable energy sources, such as solar photovoltaic (PV) systems and heat pumps (HP). However, this transition presents considerable challenges in terms of monitoring and management, as it results in a highly distributed production network that complicates national grid operations. Areas characterized by dispersed populations and a high proportion of residential buildings often encounter particular difficulties during this transition. While solar PV systems are the primary choice for domestic renewable energy, single-family houses are likely to feed a substantial portion of their production into the grid, in contrast to apartment blocks where self-consumption is significantly higher.

In this context, the role of nuclear energy is being reconsidered, particularly with the advent of Small Modular Reactors (SMRs). The present study focuses on the north-western part of the province of Varese, located in the Lombardy region of Italy, where a hybrid system based on the integration of SMRs into a solar-powered grid is being evaluated.

Figure 1: *Target geographical region for the project.*

The integration of Small Modular Reactors (SMRs) has the potential to offer a stable and reliable energy source, complementing the intermittent nature of renewable energy sources such as solar photovoltaic (PV) systems. This integration could also address the challenges associated with grid management and energy distribution in areas with dispersed populations and a high proportion of residential buildings. By providing a consistent energy supply, SMRs can help mitigate the issues arising from the highly distributed production network characteristic of renewable energy adoption in the residential sector.

1.2 Case Study

The present study assesses the technical and economic feasibility of integrating Small Modular Reactors (SMRs) into the residential sector, taking into account the specific characteristics of the area and the potential benefits and challenges associated with such an implementation.

An Italian energy company, specializing in electricity production, has been commissioned by the national government to develop a strategic project aimed at supporting the energy transition and decarbonization. This project entails the implementation of a hybrid system that combines nuclear power, renewable energy sources, and the production of hydrogen and synthetic CO_2-neutral methane. The inclusion of hydrogen production is primarily intended to decarbonize the land transport sector in the region, while the production of synthetic methane is designed to meet the residual demand for fossil fuels (e.g., heating) with CO_2-neutral gas. The project is being developed in accordance with European Union guidelines and is part of the EU-sponsored "net-zero" project cluster.

The proposed system comprises two Small Modular Reactors (SMRs), each with a nominal power output of 180 MWe, integrated with a secondary system capable of producing hydrogen through high-temperature steam electrolysis (HTSE). This system also includes the production of synthetic gas via CO_2 methanation, which is fed into the gas mains and powers a small gas turbine with a nominal electrical output of 47 MWe. The electricity generated by this integrated system, in conjunction with residential photovoltaic (PV) production, is designed to meet the entire demand of the service area. Furthermore, the plant is configured to manage demand peaks and enhance profitability and adaptability to various energy market requirements. The integrated system is designed to ensure a reliable and sustainable energy supply, addressing both the base load and peak demand requirements of the service area.

Figure 2: *Simple overview of the proposed hybrid plant integrating SMRs, hydrogen production, methanation, and residential PV.*

1.2.1 Reactor

The proposed plant comprises two Small Modular Reactors (SMRs) intended for installation in Ispra, the site of the Joint Research Center (JRC). The JRC, a research facility operated by the European Commission, was selected due to its status as an already nuclearized site, which suggests a potentially streamlined pathway for the construction of a future nuclear plant.

Furthermore, the facility houses the ESSOR reactor, which is currently undergoing decommissioning. Technically, the installation of at least one of the two SMRs could be accommodated within the existing ESSOR reactor building. The containment structure of the ESSOR reactor was oversized to accommodate its specific use case, which includes primary and secondary circuits, as well as several laboratories, all housed within the main building.

1.2.2 Secondary Plant

The secondary plant is engineered for dual-purpose hydrogen utilization. Its primary function is the production of hydrogen through high-temperature steam electrolysis (HTSE), with a consistent output designated for market supply. Additionally, the plant employs the remaining hydrogen to generate synthetic gas via CO_2 methanation. This synthetic gas production serves two principal purposes: it powers the gas turbine and creates a surplus that can be stored, thereby reducing the region's reliance on fossil fuels and supporting decarbonization initiatives.

1.3 Objectives

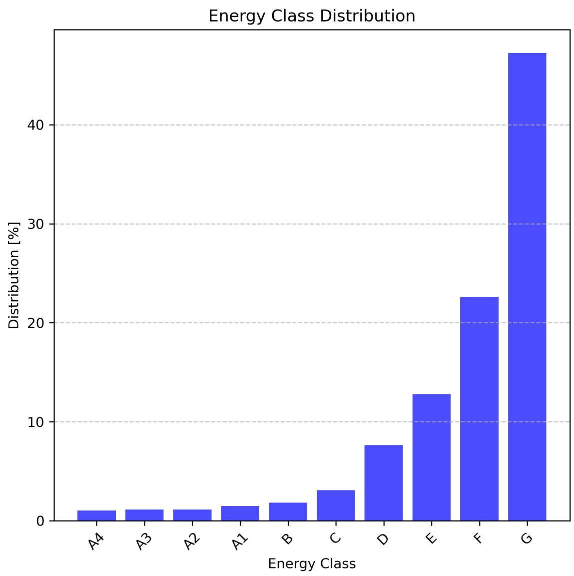
The study aims to assess both the technical and economic feasibility of the proposed installation.

From a technical standpoint, the system's ability to meet energy demands and adapt to transient conditions will be evaluated. From an economic perspective, the total plant costs will be analyzed to determine the conditions necessary for achieving a positive net present value (NPV) within a reasonable timeframe.

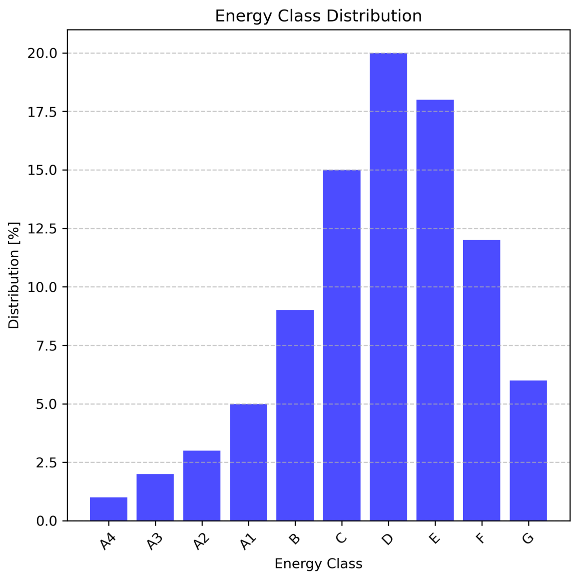
The evaluation will encompass the entire plant's operation over a 60-year period. The evolution of residential energy and building performance is simplified into four scenarios, each representative of a 15-year period:

- Years 1-15: Current state of the residential sector in the area
- Years 16-30: Significant share of fossil fuel heating systems remains in the residential sector
- Years 31-45: Balanced mix of electrification and fossil fuel systems
- Years 46-60: Residential sector is predominantly electrified with highly efficient homes

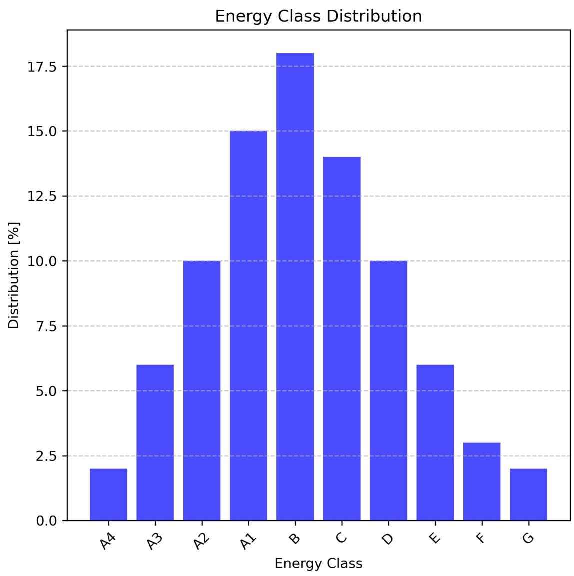
Given the constraints of the available data, the assessment will initially consider the plant to be operational as of the current year (2025). At the conclusion of the study, a more realistic starting time will be considered. The approach involves abstracting from a specific starting time by evaluating capital and operational costs in relation to each other.



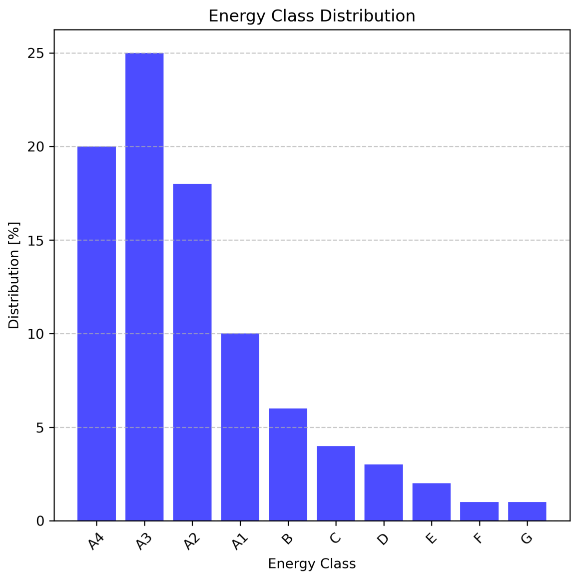
(a) 1-15 Years Scenario



(b) 16-30 Years Scenario



(c) 31-45 Years Scenario



(d) 46-60 Years Scenario

Figure 3: *Distribution of energy classes in the residential sector for the four scenarios.*

The effectiveness of the hybrid system is evaluated based on the following key aspects:

- Assessment of the investment required to ensure the plant's profitability
- The decarbonization impact
- The effectiveness of integration of the several components of the plant

1.3.1 Decarbonization Impact

The decarbonization impact will be assessed by analyzing the reduction in CO2CO2 emissions achieved through the implementation of the hybrid system. A comparative approach will be used, examining emissions from both the current residential sector energy mix and those projected for the proposed system across various scenarios.

The reduction in emissions will be quantified based on two primary factors:

- Volume of conventional gas replaced by CO_2-neutral synthetic gas

- Changes in the overall energy mix composition

Both aspects will be evaluated using grams of CO_2-equivalent per kilowatt-hour ($\frac{gCO_2-eq}{kWh}$) as the measurement unit. The decarbonization impact will ultimately be expressed as a percentage variation from the current state of the energy mix.

2 DATA COLLECTION

The foundation of this study lies in the necessity to establish a realistic energy demand profile for the residential sector within the examined region. Given the focus on residential energy consumption, the total energy demand is conceptualized as the sum of heating and cooling contributions, in addition to electricity consumption for appliances and lighting. Throughout the subsequent analysis, several simplifications are employed. However, potential improvements that could be made but were deemed beyond the scope of this work are duly noted.

2.1 Approach

2.1.1 Introduction

In Italy, the Attestato di Prestazione Energetica (APE) serves as a certification document that evaluates the energy performance of a property unit, referred to as *unità immobiliare*. This certification, which must be issued by a qualified technician, is a mandatory requirement for all buildings that are sold, rented, or undergo refurbishment.

The APE categorizes buildings into energy classes, denoted by letter grades ranging from A4 to G, which indicate the energy performance of the building. In this classification, A4 represents the highest level of efficiency, while G signifies the lowest. The certificate encompasses a wide range of information about the building, including details on geometry, mechanical and electrical systems, construction materials, and the building's intended use (e.g., commercial, residential, industrial, public, etc.).

Each unit pertains to a distinct and identifiable segment of a real estate property, such as an apartment, a house, or a commercial space, which can be individually owned, sold, or rented. For example, a building comprising three apartments will have three separate APEs, each corresponding to an individual apartment. Consequently, it is reasonable to assume that each unit serves as the residence of a single family.

2.1.2 Methodology

The objective is to ascertain the electricity demand profile for the region, denoted as $P_{e,\text{tot}}$ based on the distribution of energy classes. To achieve this, it is essential to first establish the reference demand profile for each energy class, represented as $P_{e,\text{ref},i}$, where i signifies the specific energy class.

For each energy class, it is necessary to determine the following parameters:

- Heating thermal demand $\dot{Q}_h$
- Cooling thermal demand $\dot{Q}_c$
- Domestic hot water thermal demand $\dot{Q}_{dhw}$

Additionally, the share of heat pumps, denoted as $\chi_{HP,i}$ must be established for each energy class to distinguish between electrified and fossil fuel-based heat generation. The primary energy demand, whether for electricity or fossil fuels, is calculated using average efficiencies in a simplified manner.

Regarding the electricity production from photovoltaic systems, it is essential to determine the share of buildings equipped with such systems for each energy class, denoted as $\chi_{PV,i}$. Additionally, the average size of the photovoltaic system, represented as $P^+_{e,i}$, must be established for each energy class. This parameter is further differentiated between households with and without a heat pump. A reference profile of photovoltaic (PV) production for the region is obtained and subsequently weighted according to the simulated scenario.

An additional electricity contribution arises from appliances and lighting, denoted as $P_{e,other}$, which is assumed to be consistent across all energy classes. The reference demand profile for each energy class i can then be determined using the following equation:

$$
\begin{aligned}
P_{e,\text{ref},i} = \Bigg[& \chi_{\text{HP},i} \cdot \left(\frac{\dot{Q}_{h,i} + \dot{Q}_{DHW}}{\text{COP}} + \chi_{\text{cooling}} \cdot \frac{\dot{Q}_{c,i}}{\text{EER}} \right) \\
& (1 - \chi_{\text{HP},i}) \cdot \left(\frac{\dot{Q}_{h,i} + \dot{Q}_{DHW}}{\eta_{\text{fossil}}} \right) \\
& - \chi_{\text{PV},i} \cdot \left(\chi_{\text{HP},i} \cdot P_{\text{PV,w/HP}} + (1 - \chi_{\text{HP},i}) \cdot P_{\text{PV,w/oHP}} \right) \\
& + P_{e,\text{other}} \Bigg]
\end{aligned}
\tag{1}
$$

Subsequently, the reference demand profile can be multiplied by the number of buildings in the region, denoted as N_{UI}, and the share of each energy class, represented by χ_i, to calculate the total demand $P_{e,\text{tot}}$. This relationship is expressed as follows:

$$
P_{e,\text{tot}} = \sum_{i=\text{A4}}^{\text{G}} N_{\text{UI}} \cdot \chi_i \cdot P_{e,\text{ref, i}} \tag{2}
$$

Data Sources Data pertaining to APEs for the relevant municipalities within the region were extracted from the CENED database [2]. The selection of municipalities was based on the map of primary substations provided by the national grid operator (GSE) [3]. The dataset was subsequently filtered to include exclusively residential buildings, thereby excluding commercial and industrial structures. Additionally, data points deemed unhealthy were filtered out at each stage of the process, primarily by considering only the 99th percentile. This filtering was necessitated by the extensive volume of data contained within the complete regional database. Following this preliminary filtering, a more manageable CSV file was obtained. The number of buildings was calculated using data from the ISTAT database [4], where the population of the municipalities was extracted and divided by the average household size in the province.

2.2 Heating and Cooling Demand

For each energy class, a representative residential building from the region under study was selected, and the hourly thermal demand for heating and cooling was computed over the course of a year. This computation was performed using professional software (TERMOLOG), which enables dynamic simulation of a building's energy system. The calculations were conducted in accordance with the UNI EN ISO 52016 standard. A single reference building with a consistent utilization profile was employed for each energy class. The result of these calculations is an hourly profile of the thermal demand for both heating and cooling.

2.3 Domestic Hot Water Demand

The domestic hot water demand is assumed to be uniform across all energy classes. It was computed based on the consumption profile illustrated in [5], with the Swiss profile selected as a reference due to its similarity to the region under study. This demand was then scaled according to the average number of people per household in the region, which is 2.25 [4]. The resulting thermal demand for domestic hot water was subsequently added to the heating demand profile.

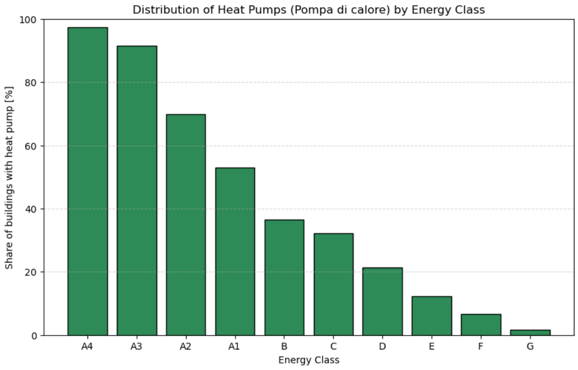
2.3.1 Normalization of the Data

Given that the data is based on real buildings, normalization was necessary to obtain a reference value for each energy class. This normalization process considered the useful heated area of the building and the EP,nren (Primary Energy Demand for non-renewable energy) value from the APE. The EP,nren is the metric used to determine the energy class of a building and is defined as the energy consumption per unit of area per year. By multiplying the EP,nren by the useful heated area of the building, the total primary energy demand for non-renewable energy was obtained. Conversion factors for each energy class were derived as the ratio between the values obtained from the averages extracted from the CENED database [2] and those of the reference buildings. This is expressed as follows:

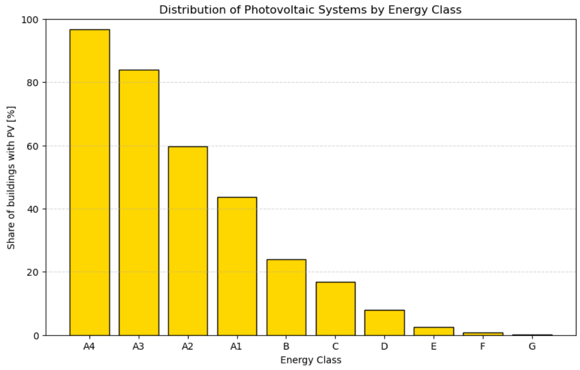
$$
C = \frac{\text{EP,nren}_{\text{CENED}} \cdot A_{\text{r,CENED}}}{\text{EP,nren}_{\text{REF}} \cdot A_{\text{r,REF}}} \tag{3}
$$

2.4 Heat Pump distribution and Cooling Demand

The share of heat pumps per energy class was identified. Additionally, the share of buildings equipped with cooling systems was determined, which currently stands at 9.58%. This relatively low percentage is expected, given that the region under investigation experiences mild summer temperatures.



(a) *Distribution of heat pumps across energy classes.*

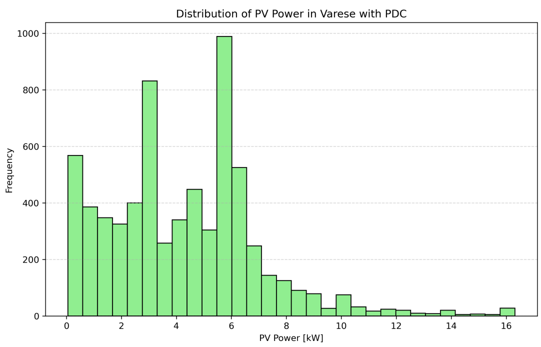


(b) *Distribution of photovoltaic systems across energy classes.*

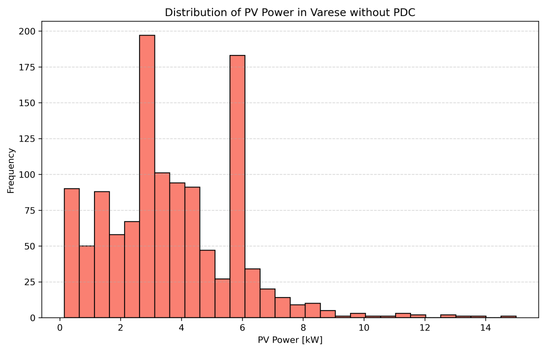
Figure 4: *Distribution of heat pumps and photovoltaic systems across energy classes.*

2.4.1 Photovoltaic System Size

Furthermore, the data was divided into two groups: buildings with heat pumps and those without. The distribution of the size of the installed photovoltaic systems was then plotted for each group. The average size, calculated based on the 99th percentile, was determined for both groups



(a) *Photovoltaic system size distribution for buildings with heat pumps.*



(b) *Photovoltaic system size distribution for buildings without heat pumps.*

Figure 5: *Distribution of photovoltaic system sizes based on heat pump presence.*

The average values for the size of the photovoltaic systems are 4.30 kW for buildings with heat pumps and 3.70 kW for those without heat pumps.

2.4.2 Reference Photovoltaic Production

To establish a reference photovoltaic (PV) production profile representative of the region under study, the list of municipalities was first input into the Nominatim API [6] to obtain the coordinates of the center of each municipality. Subsequently, the PVGIS API [7] was utilized to acquire the hourly production profile for each municipality. The angle was fixed at an average of $22°$, and a plant size of 11 kW was assumed. For each municipality, a profile was generated for every $10°$ of the azimuth angle, ranging from $0°$ to $360°$. These profiles were then weighted according to a Gaussian distribution to reflect the preference for orienting photovoltaic systems towards the south. This process yielded a reference photovoltaic production profile for each municipality, denoted as $P^{+}_{e,m}$, where m represents the municipality. The reference production profile for each energy class, $P^{+}_{e,ref}$, was then obtained by weighting the production profile of each municipality based on its population [4].

2.5 Other Electricity Demand

The electrical demand for appliances and lighting is modeled uniformly across all energy classes. The demand profile was manually defined on an hourly basis for an average day, with distinctions made between the following scenarios:

- Winter Weekday
- Winter Weekend
- Summer Weekday
- Summer Weekend
- Winter Holiday
- Summer Holiday

Additionally, an absence factor was incorporated to account for periods when a significant number of residents are on holiday, particularly during the summer.

The profile was established by considering the following contributions:

- standby and always on appliances
- lighting
- appliances

During the summer, the need for cooling fans was included in the model. Furthermore, a parameter was introduced to represent the share of buildings equipped with induction cooking systems.

2.6 Validation

To validate the acceptability of the proposed method, the total demand was compared with data available online. Specifically, it was compared to the average annual demand in the residential sector for the province of Varese [8], which was reported to be 19311931 kWh in 2023. This value reflects a decreasing trend in recent years, with figures of $2016kWh$ in 2022 and $2119kWh$ in 2021.

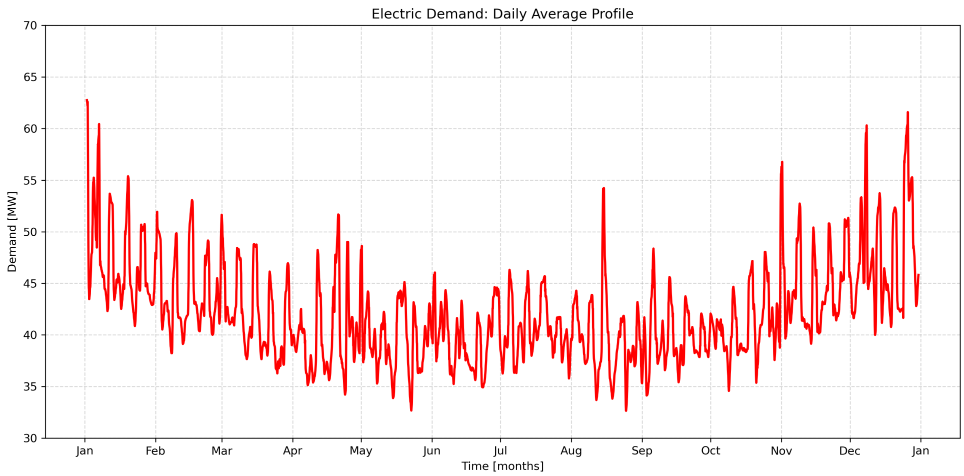
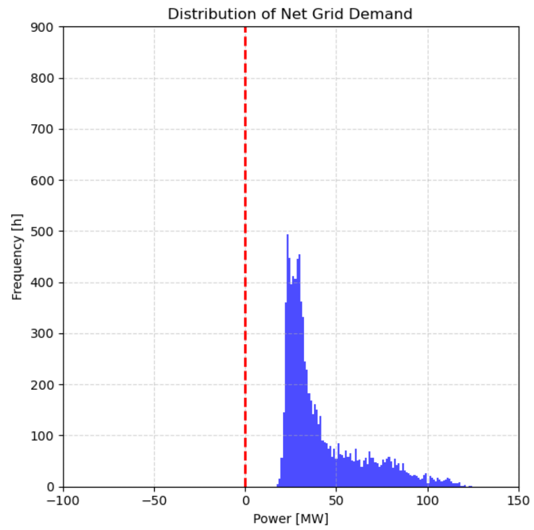


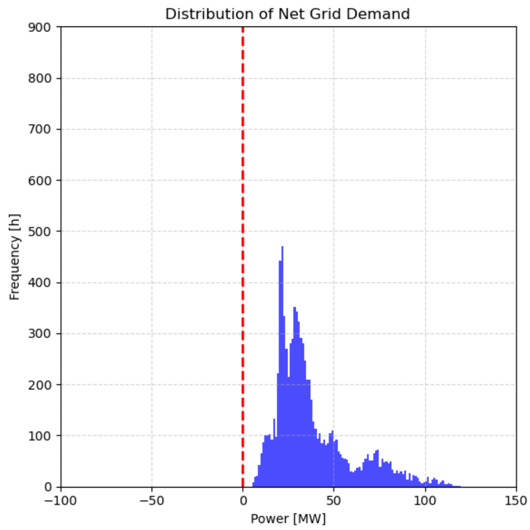
Figure 6: *Daily averages thoughout the year of the resulting electric demand profile considering today's energy class distribtion.*

2.7 Demand Distribution

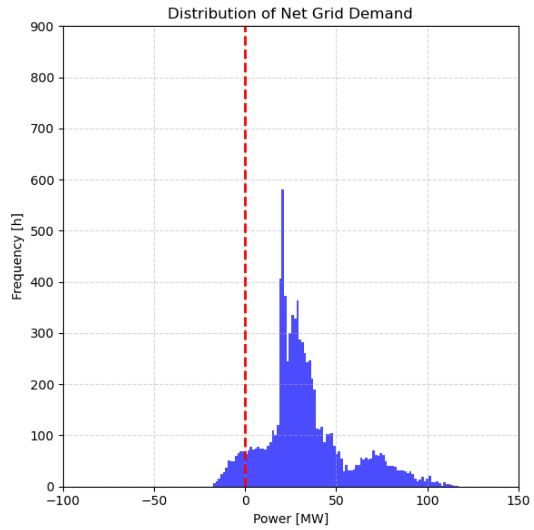
To enhance the visualization of demand behavior, the hourly distribution of demand was plotted for each scenario. This graphical representation aids in understanding the temporal variations and patterns in energy consumption across different scenarios.



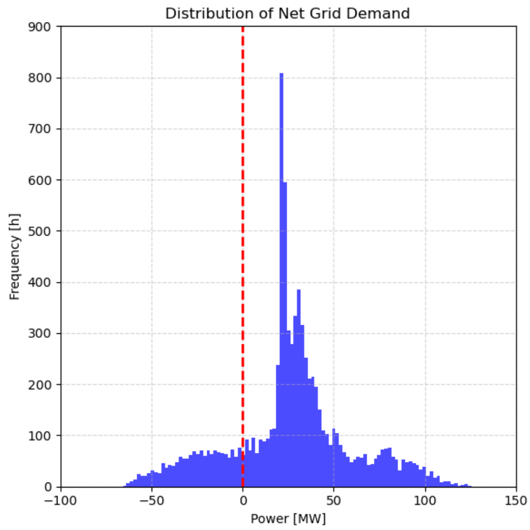
(a) *Today-15 Years*



(b) *16-30 Years*



(c) *31-45 Years*



(d) *46-60 Years*

Figure 7: *Hourly Distribution of the Electrical Demand in the Four Scenarios.*

The plots provide a clear depiction of the increase in the gap between minimum and maximum demand as well as the increase in photovoltaic excess production over the years.

3 TECHNICAL PLANT MODELLING

The TANDEM project library [9] was widely used to model the entire plant, with the exception of the methanation reactor, which was modeled from scratch in Dymola. The entire plant was modeled using Dymola, a technical software that employs the Modelica language for the dynamic analysis of complex systems. To satisfy the overall energy demand of the target region, the plant design integrates two 177 MWe Small Modular Reactors (SMRs) with fifteen High Temperature Steam Electrolyzers (HTSEs) and a Methanation Reactor. The Methanation Reactor produces nearly CO_2-neutral syngas for a small Gas Turbine with a capacity of 47 MWe. The TANDEM project library [9] was extensively utilized to model the entire plant, with the exception of the Methanation Reactor, which was modeled from scratch in Dymola.

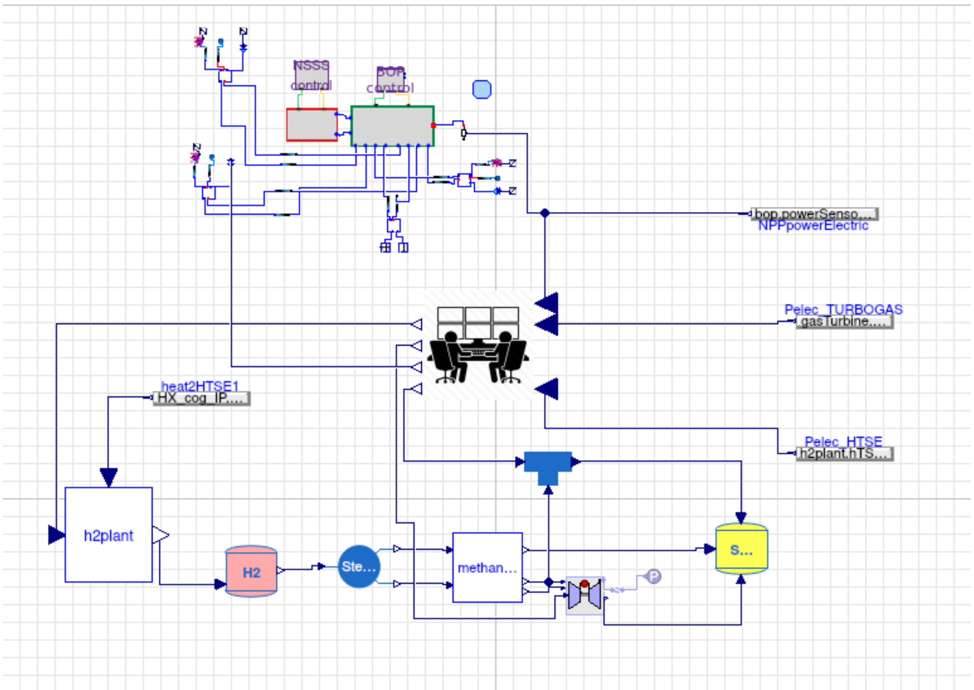


Figure 8: *Dymola model of the proposed hybrid plant.*

3.1 Main assumptions and plant layout

The number of reactors and High Temperature Steam Electrolyzer (HTSE) modules is determined based on the following considerations:

- The minimum number of installed HTSE modules must be sufficient to sustain the Gas Turbine operation at nominal power, ensuring adequate methane production.
- The maximum number of installed HTSE modules is constrained by the overall energy excess of the system during the least demanding hour of the year.
- The overall energy excess of the system is dependent on the number of Small Modular Reactors (SMRs).

To determine the optimal number of Small Modular Reactors (SMRs) and High Temperature Steam Electrolyzer (HTSE) modules that would minimize the overall plant cost while meeting energy demand, an optimization problem would typically be required. However, due to the complexity of the problem, such an optimization is not performed in this study. Specifically, establishing a direct and accurate functional relationship between the plant's actual revenues and the number of components would necessitate a highly detailed model, as well as a comprehensive set of equations to represent all operational constraints. Developing and solving such a model would be computationally intensive and disproportionate to the scope and objectives of the current project.

The number of High Temperature Steam Electrolyzer (HTSE) modules was determined by taking the average between the minimum number required (12) and the maximum allowed (18, with two Small Modular Reactors). This calculation resulted in a total of 15 HTSE modules. Given that HTSE modules require a substantial amount of electricity to operate, the number of Small Modular Reactors (SMRs) was set to two. This configuration enables efficient operation of the entire plant, with the need to import electricity for less than 10% of the hours in a year.

To reduce the computational burden of the simulation in Dymola, the system is modeled using a single Small Modular Reactor (SMR) and a single High Temperature Steam Electrolyzer (HTSE) module. The actual electricity output from the SMR and hydrogen production from the HTSE are then scaled by multiplying the simulated values by the respective number of units in the actual plant. These scaled values are used to evaluate the overall energy balance and to feed the syngas production module.

The Small Modular Reactors (SMRs) supply electricity to the grid, as well as both electricity and the required heat to the hydrogen production unit. The High Temperature Steam Electrolyzers (HTSEs) generate hydrogen from water using the thermal and electrical energy provided by the SMRs.

At each time step, 10% of the produced hydrogen is temporarily stored in a buffer for future sale, while the remaining 90% is sent to the methanation reactor, where it reacts with carbon dioxide to produce methane. The profile and magnitude of future hydrogen demand are unknown; the simplified assumption is made to facilitate the technical evaluation of the plant's operation while considering a reasonable amount of hydrogen to be stored and sold. Excessive hydrogen storage would not be feasible in a real-world scenario due to the significant investments required in storage infrastructure.

The syngas produced by the methanation reactor is used as fuel in the gas turbine, with any surplus stored in a buffer tank. The stored syngas can later be utilized to cover the residual fossil fuel demand of the region, such as the remaining heat demand not met by other sources, thereby substantially contributing to the decarbonization of the energy mix.

3.1.1 SMR

The TANDEM library [9] was utilized to model the Small Modular Reactor (SMR), employing a simplified yet detailed representation of the reactor. This model includes various components that enable a realistic simulation of the reactor's behavior, capturing the dynamics of the primary and secondary loops, turbines, condenser, and associated control systems.

The selected model is part of the POLIMI section of the SMR branch in the TANDEM library. Specifically, the `Test_BOPfluid_NSSS` [9] is a test model of the 177 MWe light-water cooled SMR, based on the European SMR (E-SMR) design as proposed by the Euratom ELSMOR project [10]. The model is constructed using elementary components from the ThermoPower library [9].

The employed test model was initially designed for cogeneration, providing both electricity and thermal energy. To facilitate this dual purpose, the Balance of Plant includes three steam extraction points: High Pressure (HP), Intermediate Pressure (IP), and Low Pressure (LP). These extraction points enable heat to be drawn from different stages of the turbine. In the current configuration, the heat required for the High Temperature Steam Electrolyzer (HTSE) is extracted via the IP tap, located immediately downstream of the HP turbine stages. The HP and LP extraction lines are disabled and manually set to zero, as they are not utilized in this setup.

For the purpose of this study, the detailed modeling of the primary loop is not discussed, as the focus is on the secondary loop and its interaction with the HTSE modules. However, a detailed explanation of the reactor's core and primary loop modeling can be found in the TANDEM library documentation [10].

SMR specifics

- Core Thermal Power: 540 MW_{th}
- Core Inlet Temperature: 300 °C
- Core Outlet Temperature: 324.5 °C
- Primary Loop Pressure: 150 bar
- Nominal Coolant Mass Flow Rate: 3700 kg/s
- Thermodynamic Efficiency: 0.33
- Electrical Power Output: 177 MW_e
- Secondary loop: Rankine cycle with multiple turbine stages
- Steam extraction points pressure and temperature [10]:
  - HP: 45 bar, 240 °C
  - IP: 7.5 bar, 150 °C
  - LP: 0.8 bar, 80 °C

3.1.2 H2 Plant

The hydrogen production plant is modeled using the TANDEM library [9], which offers a detailed representation of a High Temperature Steam Electrolyzer (HTSE) based on the ThermoSysPro library. This Solid Oxide Electrolyser Cell operates at high temperatures and is designed to produce hydrogen from water by utilizing both thermal and electrical energy supplied by the Small Modular Reactor (SMR). The model captures the dynamic behavior of the HTSE and simulates the hydrogen production process accordingly.

Water goes through several heat-exchangers to reach the target temperature of 750 °C as steam:

- A pre-heater.

- A boiler, fed by the tap-steam extracted from the SMR BOP (at 150 °C).
- A second pre-heater, fed by the recovery heat from unconverted steam.
- An electrical heater.

Each electrolyzer stack utilizes electricity to act on high-temperature steam and produce hydrogen. The model receives as input a target hydrogen production rate, which is further defined in the demand section, and the thermal energy extracted from the Small Modular Reactor's (SMR) steam tapping point.

The electricity consumption is related to the hydrogen production rate according to the relationship:

$$\dot{m_{H2}} = \frac{P_{e,available}}{214} + 0.197775 \tag{4}$$

This relationship was derived by running a sweep simulation of the High Temperature Steam Electrolyzer (HTSE) model in Dymola.

A key modeling assumption is that the heat input to the HTSE corresponds to the actual value of the heat exchanged in the ideal Heat Exchanger of the Intermediate Pressure (IP) tap loop, once the tap-steam flow rate is defined.

Figure 9: *Simple schematic of the Intermediate Pressure (IP) tap loop.*

The tap-steam flow rate is defined in the Control Room section, where the setpoint for steam extraction is calculated based on the target hydrogen (H_2) production. The relationship between the tap-steam flow rate and the hydrogen production rate is determined by running a sweep simulation of the High Temperature Steam Electrolyzer (HTSE) model in Dymola.

Based on the thermal input and several operational parameters, the HTSE model calculates the corresponding hydrogen production, resulting in a computed mass flow rate of H_2. The estimate of the energy consumption associated with hydrogen production is also computed within the model and subsequently included in the overall energy balance of the system.

HTSE specifics

- Operating Temperature: 750 °C
- Electricity Consumption: related to the hydrogen production rate
- Nominal Production Rate: 0.1 kg/s of H_2 per module
- Number of Modules: 15

The hydrogen (H_2) plant is equipped with a buffer tank used to temporarily store the portion of hydrogen intended for sale, while the remainder is sent directly to the methanation reactor.

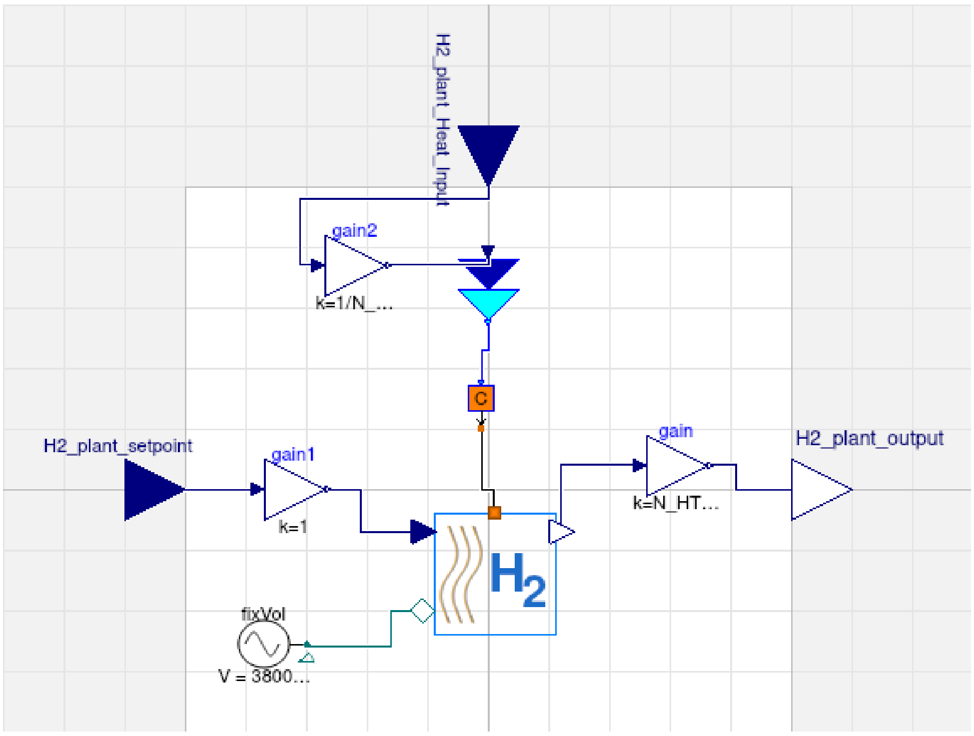


Figure 10: *Simple schematic of the H_2 plant model in Dymola.*

As the model is designed to simulate the operation of a single High Temperature Steam Electrolyzer (HTSE) module, the target hydrogen production rate (and the effective output production rate) is obtained by dividing (and multiplying) the total input (and output) value by the number of modules (15), through a simple "Gain" component.

3.1.3 Methanation Reactor Modeling Approach

The methanation reactor was modeled to simulate the conversion of hydrogen (H_2) and carbon dioxide (CO_2) into methane (CH_4) and water (H_2O) through the Sabatier reaction:

$$\mathrm{CO_2} + 4\mathrm{H_2} \rightarrow \mathrm{CH_4} + 2\mathrm{H_2O} \qquad \Delta H = -165.1\ \mathrm{kJ/mol} \tag{5}$$

To estimate the amount of CH_4 produced from a given reactant feed, a conversion efficiency, η_{chem}, was introduced. This parameter, with a default value of 0.78, represents the maximum chemical conversion efficiency of catalytic methanation, as reported by Götz et al. [11]. It defines the fraction of the limiting reactant (either CO_2 or H_2) that is effectively converted into CH_4, accounting for real-world inefficiencies and the incomplete conversion typical under practical temperature and pressure conditions.

The molar output flows are then determined based on reaction stoichiometry:

$$\begin{aligned}
\dot{n}_{\mathrm{CH_4}} &= \dot{n}_{\mathrm{lim}} \cdot \eta_{chem} \\
\dot{n}_{\mathrm{H_2}O} &= 2 \cdot \dot{n}_{\mathrm{CH_4}} \\
\dot{n}_{\mathrm{H_2}} &= \dot{n}_{\mathrm{H_2,in}} - 4 \cdot \dot{n}_{\mathrm{CH_4}} \\
\dot{n}_{\mathrm{CO_2}} &= \dot{n}_{\mathrm{CO_2,in}} - \dot{n}_{\mathrm{CH_4}}
\end{aligned}$$

The model is reported in the appendix, along with the other components developed from scratch in Dymola.

Stoichiometry and Output Mixture Given that the reaction is not ideal, adhering to the stoichiometric ratio (1:4) would result in a gas mixture with a lower heating value (LHV) compared to natural gas. To enhance the energetic and economic value of the product, the system operates with an excess of hydrogen, using a molar ratio of 1:8 between CO_2 and H_2. This results in an output gas mixture with the following approximate molar composition:

Component	Molar Fraction [%]
CH_4	41.1
H_2	30.7
CO_2	28.2

Table 1: *Approximate molar composition of the output gas mixture.*

Water is assumed to be completely removed from the product stream. The resulting lower heating value (LHV) of the dry gas mixture is 57.4359 MJ/kg.

CO_2 **Usage** Based on the product gas composition, the synthetic gas mass flow rate is 2.6325 kg (reference case). Given the process stoichiometry and conversion efficiency, the corresponding input of CO_2 is 3.7 kg. From this, we deduce that producing 1 kg of synthetic gas requires approximately 1.41 kg of CO_2.

3.1.4 Gas Turbine

The gas turbine selected for this system is the GE LM6000, an aeroderivative machine capable of burning fuel with high hydrogen fractions and rapidly adjusting its power output. This makes it a valuable component for hybrid energy systems with high renewable penetration. The GE LM6000 is modeled by modifying the "GasTurbineSimplified" component found in the ThermoPower section of the TANDEM library [9].

The original ThermoPower model is a generic gas turbine model, which was adapted to represent the GE LM6000 by scaling the relevant parameters by a factor of $1/5$ to match the LM6000's nominal specifications. The ThermoPower model initially only had a power level input, so additional inputs and outputs were incorporated to fulfill its role in the Sette Laghi model.

GE LM6000 specifics

- Nominal Power: 47 MW_e
- Efficiency: 0.4
- Fuel: Syngas (Methane and Hydrogen)
- Mass Flow Rate of Syngas: 2.42 kg/s
- Maximum Ramp Rate : 30 MW/min
- Minimum Load: 35% of the nominal power

The gas turbine model takes as inputs the fuel heating values and the power load setpoint, which is computed in the Control Room, to calculate the required mass flow rate of syngas. This information is then communicated to the Syngas Buffer. Since the detailed modeling of the combustion process is not the focus of this study, the model is simplified to concentrate on the mass flow rate of syngas required to meet the setpoint load. Although flue gases contain valuable energy content, they are not modeled in this study, as the primary focus is on syngas production.

The inputs of the Gas Turbine are:

- $turbine_load$: Setpoint for the power level.
- LHV: Lower heating value of the syngas.
- HHV: Higher heating value of the syngas.

The only output of the Gas Turbine is:

- $fuelFlowRateOut$: the mass flow rate of syngas required to meet the setpoint load

3.1.5 H_2 and CH_4 buffers

The hydrogen (H_2) and methane (CH_4) buffers are designed to temporarily store the produced hydrogen and methane, respectively. These components are modeled using a simple mass balance equation.

- The H_2 buffer receives as input the mass flow rate of the produced hydrogen from the HTSE and sends as output the 90% of the produced hydrogen to the methanation reactor.
- The CH_4 buffer receives as input both the mass flow rate of the produced syngas from the methanation reactor and the mass flow rate required by the gas turbine to guarantee its setpoint load. The turbine's requested mass flow rate is an output of the turbine's model itself.

The instantaneous content of each buffer is then computed through a mass balance equation:

$$\frac{d}{dt}\text{Content} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} \tag{6}$$

As stated previously, the H_2 buffer is used to store the 10% of the produced hydrogen, which is then sold. The CH_4 buffer is used to store the syngas produced by the methanation reactor, which can later be used to cover the residual fossil fuel demand of the region.

3.1.6 Control Room

The Control Room is the central component of the plant model, responsible for managing the overall operation of the system. It defines the interconnections between the various components, such as the SMR, HTSE, Methanation Reactor, and Gas Turbine.

The Control Room takes as inputs:

- P_{nuc}: the hourly electrical power produced by the SMR.
- $P_{turbine}$: the hourly electrical power produced by the Gas Turbine.
- $Demand$: the hourly net grid demand, considering the contribution of the PV production.
- P_{HTSE}: the hourly electrical power consumed by the HTSE modules.

The Control Room calculates the Energy Balance of the system using the following equation:

$$\text{EnergyExcess} = n_{\text{reactors}} \cdot P_{\text{nuc}} + P_{\text{turbine}} - P_{\text{HTSE}} \cdot n_{\text{HTSE}} - \text{Demand} \quad [\text{MW}] \tag{7}$$

The outputs of the Control Room are:

- $setpoint_{H_2}$: Setpoint for the hydrogen production rate.
- $turbine_load$: Setpoint for the turbine load.
- $mflow_tapSteam$: Setpoint for the mass flow rate of steam extracted from the SMR's intermediate pressure tap.

Within the Control Room, the setpoints for the various components are imported as .csv files. These setpoints are generated beforehand by a Python script focused on analyzing single-day demand profiles, as described in the following section. The Control Room also computes the instantaneous energy excess based on the Energy Balance Equation, which is useful for economic assessment.

4 METHODOLOGY

Using the Dassault Systèmes Dymola software [12], a production model was developed based on the object-oriented Modelica language. Four representative scenarios were evaluated, as described in Section 1, each representing a 15-year operational phase of the plant, covering a total time span of 60 years. The system is assumed to begin operation in 2025 and continue until 2085.

The optimal time horizon for dynamic simulation in this framework is set to 24 hours to capture the daily variations in energy demand and the system's response to them.

The simulation methodology follows these steps:

1. For each scenario, an annual net electricity demand profile for the regional grid was derived through the data analysis developed in Section 2.
2. The annual profile was then clustered into five groups of days with similar load characteristics. These representative days were then dynamically simulated with Dymola.
3. A Python-based algorithm reconstructs the full annual profile by substituting actual days with the corresponding simulated cluster days.

4. Monthly energy balances, including electricity sold to the community, surplus energy exported to the market, and energy deficits covered by purchases from the market, are extracted and used as input for the economic analysis of the plant.
5. The yearly hydrogen and syngas production data are also extracted from the simulation results and added as input for the economic analysis.

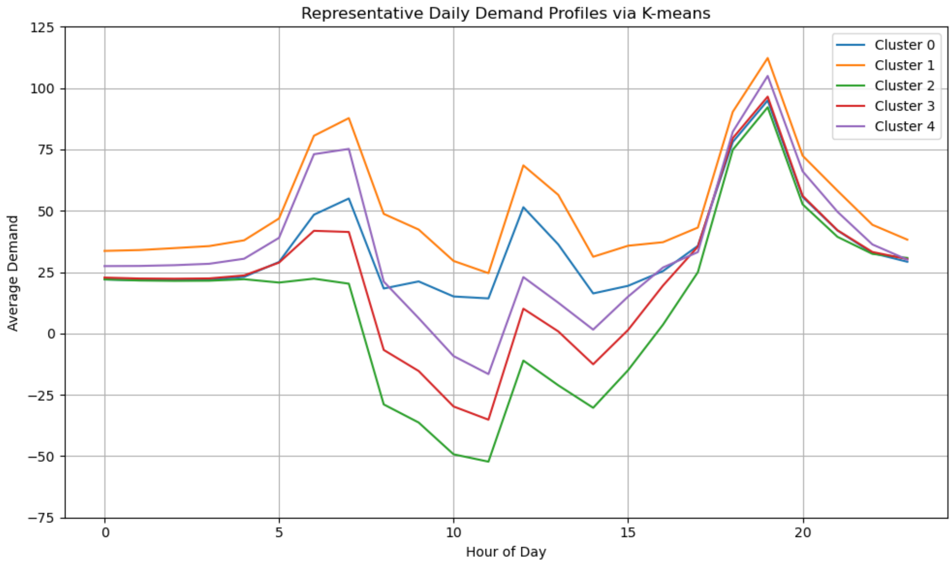


Figure 11: *Example of clusters for an annual demand profile.*

4.1 Dymola simulation features

The Dymola model simulates the dynamic behavior of the plant over a 24-hour period but requires pre-processed input data to accurately account for all relevant phenomena. The pre-processing methodology described below is based on the single-day demand profiles of the clusters previously obtained.

4.1.1 Model inputs

As highlighted in the Control Room subsection 3.1.6, the model requires, in addition to the hourly net grid demand (i.e., one of the cluster profiles), a set of daily setpoints for the following variables:

- Turbine load, constrained between 35% and 100% of the maximum load.
- HTSE load, constrained between 60% and 100% of the maximum load.
- Steam tap flow from the intermediate pressure stage of the SMR.

Both the turbine and High Temperature Steam Electrolyzer (HTSE) loads are subject to lower operational limits to prevent the risk of damaging the components when operated below their minimum thresholds.

4.1.2 Turbine and HTSE load setpoints

To evaluate the turbine and High Temperature Steam Electrolyzer (HTSE) load setpoints, the proposed data pre-processing model first computes the electricity available for the hydrogen plant. This calculation is based on the following assumptions:

- SMRs operate continuously at their nominal power output
- Gas turbine operates at its minimum load of 35% to ensure it never runs below its allowed operational limit

This is expressed by the equation:

$$P_{e,\text{available}} = P_{e,\text{SMR}} + P_{\text{turbogas}}(35\%) - P_{e,\text{grid}} \tag{8}$$

This calculation helps determine the load profile of the hydrogen (H_2) plant based on the load vs. electric power relationship described in the previous section. This relationship was obtained by performing a sweep simulation of the High Temperature Steam Electrolyzer (HTSE) model in Dymola.

The load profile of the hydrogen plant is given by:

$$\text{load}_{H_2} = \left(\frac{P_{e,\text{available}}}{214} + 0.197775 \right) \cdot \frac{1}{N_{\text{HTSE}} \cdot 0.1} \tag{9}$$

Where the mass flow rate of hydrogen produced, $\dot{m}_{H_2} = \frac{P_{e,\text{available}}}{214} + 0.197775$, is divided by the nominal mass flow rate produced by one module (0.1 kg/s) multiplied by the number of modules N_{HTSE}.

The HTSE load profile is then clipped to remain within the operational range of 60% to 100%, where 60% represents the lower operational limit.

This clipped HTSE load profile subsequently defines the actual electricity consumption of the H_2 plant.

Based on the updated system energy balance, the turbine load profile is then determined as follows:

$$\text{load}_{\text{turbine}} * P_{e,\text{nom,turbine}} + P_{e,\text{SMR}} = \text{HTSE}_{\text{consumption}} + P_{e,\text{grid}} \tag{10}$$

The turbine load must also be clipped. In this case, the lower limit was set to 60% instead of the physical minimum of 35%, in order to:

- Increase the overall electricity production available for sale.
- Avoid the more emission-intensive operation associated with lower load levels.

The resulting clipped turbine load profile, together with the actual consumption of the HTSE modules, determines the final electricity balance of the integrated plant.

The calculations show that, depending on the specific scenario, the system needs to import electricity from the grid in only 4% to 7% of the hours in a year, enabling highly efficient operation of the entire plant.

Ramping up and ramping down rates for both the turbine and the High Temperature Steam Electrolyzer (HTSE) were considered in the calculations.

According to the technical documentation of the selected gas turbine model, the ramping rates for the turbine are essentially unrestricted, as it can quickly reach its nominal power output.

In contrast, the HTSE module is subject to a ramping rate limit of 10% per hour. This constraint allows the model to reflect the physical limitations of the component, which cannot instantaneously adjust its load as the initial computed profile might suggest.

A Python function manages the ramping behavior of both the turbine and HTSE loads, updating the previously calculated load profiles and corresponding energy balances accordingly.

4.1.3 Steam Tap Flow Setpoint

The steam tap flow setpoint is determined based on the High Temperature Steam Electrolyzer (HTSE) load profile, which is computed as described previously.

The relationship between the tap steam flow rate and the hydrogen production rate is established by performing a sweep simulation of the HTSE model in Dymola.

4.2 Results Post-Processing

The results of the Dymola simulation are post-processed to extract the monthly energy balances of the plant, which are then used for the economic analysis. First, a `.csv` file containing the hourly simulation results is generated, and the full year is reconstructed as outlined in the previous section.

4.2.1 Energy Balances

Each annual electricity balance includes:

- Electricity sold to the community.

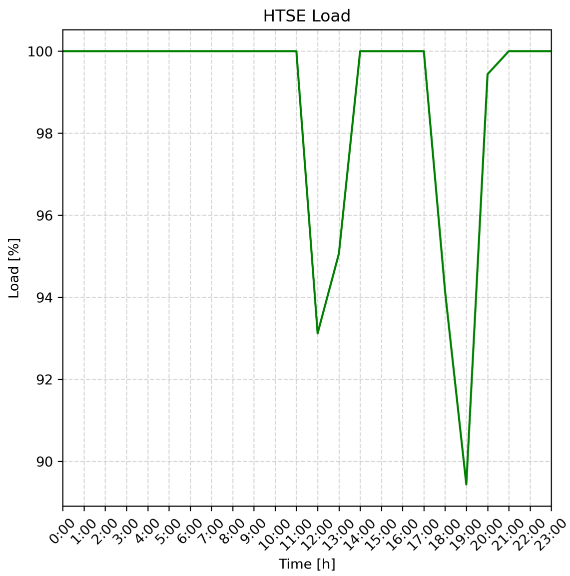


Figure 12: *Example of 1-day load profile for HTSE.*

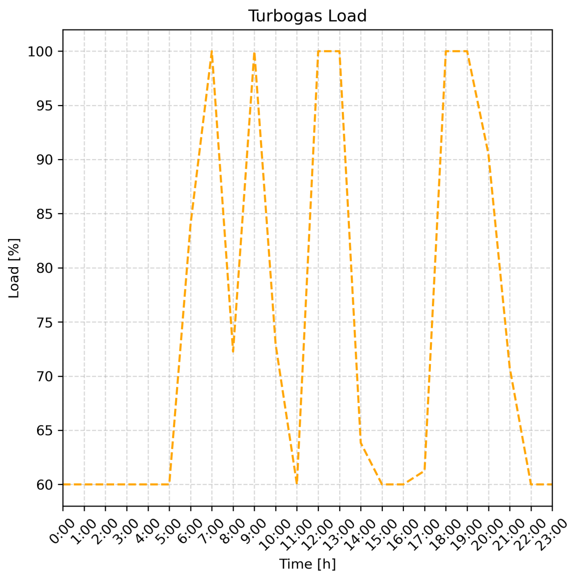


Figure 13: *Example of 1-day load profile for gas turbine.*

- Surplus energy exported to the market.
- Energy deficits covered by purchases from the market.
- Energy acquired from the community's photovoltaic excess production.

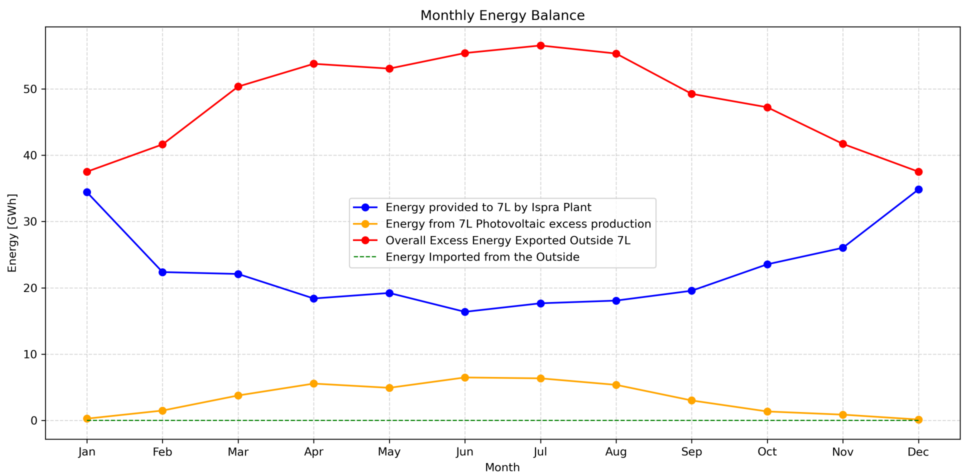


Figure 14: *1-year energy balance for the last simulated period (year 45 to 60).*

As the different scenarios progress over the plant's lifetime, a substantial increase is observed in the energy acquired from the community's photovoltaic excess production. This trend reflects the growing share of renewable energy sources in the region.

Monthly aggregate balances are then cumulatively computed to serve as input for the financial analysis.

4.2.2 Hydrogen and Syngas production

Hydrogen and syngas production data are extracted from the simulation results and used to calculate the annual product balances. Focusing on yearly balances simplifies the economic evaluation, as it assumes a stable selling price for these products throughout the year.

The yearly hydrogen production is calculated by summing the daily production values over the entire year. The total hydrogen production is expressed in energy terms (MWh), using the lower heating value (LHV) of hydrogen, which is approximately $36.11\,\frac{kWh}{kg}$.

The stored syngas is assumed to contribute towards meeting the community's remaining fossil fuel demand, thereby supporting the region's decarbonization efforts. Accordingly, the yearly syngas production is converted into energy units (MWh) using the lower heating value (LHV) of the produced gas, which is approximately $15.95\frac{kWh}{kg}$.

5 ECONOMIC STUDY

5.1 Introduction

This economic assessment evaluates both investment and operational costs for the proposed integrated energy system, covering the estimation of CAPEX and OPEX for each subsystem and calculating the expected revenues. Key financial indicators such as Net Present Value (NPV) and other metrics are determined to assess the economic viability of the solution, complemented by a sensitivity analysis to examine how variations in techno-economic parameters impact profitability and robustness.

The analysis adopts the perspective of a public-sector energy agency, acting as a government-backed institution tasked with implementing regional decarbonization strategies. This agency benefits from dedicated funding from the Italian government in the form of zero-interest loans, facilitating investment in hybrid low-carbon infrastructure.

The assessment spans a 60-year project horizon, divided into four incremental 20-year scenarios. While the analysis assumes a start year of 2025, which is not practically feasible for commissioning such complex infrastructure, this date was chosen to ensure a consistent framework amid limited long-term forecasting data. The first scenario considers the fully integrated system operational from the outset, but initial calculations suggest that such a configuration would not be economically sustainable under current market conditions. Consequently, an alternative scenario was explored, beginning with a fully electric configuration to then reach a subsequent integration of hydrogen and syngas production.

5.2 CAPEX

The estimation of capital expenditures (CAPEX) has been carried out for the main components of the proposed infrastructure, namely the E-SMRs, high-temperature steam electrolysers (HTSE), the hydrogen storage buffer, the hydrogen compressor, the gas turbine, and the methanation reactor. It is important to note that the balance of plant (BoP) costs and system integration costs have already been incorporated within the CAPEX figures reported, therefore they do not constitute separate cost items.

5.2.1 Capex for E-SMRs

The CAPEX for the E-SMRs have been estimated based on existing literature related to the TANDEM project. According to it, both reactors are assumed to have a technical lifetime of 60 years, and the specific investment cost, referred to the year 2023, is estimated at 6,050,000 EUR per MWe. This value emerges from the review conducted by CEA, where different references report investment costs ranging from 3,874,000 EUR/MWe to 7,712,000 EUR/MWe (TANDEM Project Documentation, 2023). This results in a total CAPEX of approximately 2,141.7 million EUR, considering a combined electrical output capacity of 354 MWe. This cost estimation is directly linked to the design choices adopted in the Modelica simulations, where the same libraries used within the TANDEM project framework have been selected, ensuring consistency between technical modeling and economic assessment.

5.2.2 Capex for HTSE

The CAPEX for the HTSEs have been estimated again from the literature of the TANDEM project, in line with the methodology adopted for the E-SMRs. According to the TANDEM documentation, the specific investment cost for HTSE technology is approximately 780,000 EUR per MW of nominal power. Given that the system design includes fifteen electrolyser units, each with a nominal power of 21 MW, this leads to a total CAPEX of approximately 245.7 million EUR for the HTSE section.

Each HTSE unit is assumed to have a technical lifetime of around 20 years, so that considering a 60-year project horizon, the units will need to be replaced twice after the initial investment. It is, however, reasonable to assume that the cost of HTSE technology will decrease over time as the technology matures and economies of scale are realized. The estimation of the future reduction in CAPEX relies on the cost trends observed in the hydrogen market. The report 'The European Hydrogen Market Landscape'

(2024) highlights that the production cost of green hydrogen decreased by 0.25 EUR/kg over one year, falling from 6.86 EUR/kg to 6.61 EUR/kg, corresponding to an annual reduction of approximately 3.5%. Conversely, data from the German market indicate a period of price stability (European Energy Exchange, 2025). Balancing these observations, and given the strong correlation between production costs and selling prices, an average annual cost reduction of 1.75% is plausible, leading to a decrease of approximately 30% after the first 20-year period. In the subsequent 20-year period, technological improvements will continue, albeit at a slower pace as the technology reaches maturity. Therefore, for the second replacement cycle, a more conservative annual decrease rate of 0.875% has been assumed, corresponding to a cumulative CAPEX reduction of about 15% over the second 20-year period. This anticipated decrease is primarily attributed to technological learning curves and increased manufacturing efficiencies as HTSE technology becomes more established in the market.

5.2.3 Capex for Hydrogen Storage Buffer

Also the CAPEX for the hydrogen buffer storage system have been estimated looking at the TANDEM project. In this case the specific investment cost is reported as 490 EUR per kg of hydrogen storage capacity. The hydrogen buffer has been sized to accommodate a maximum of one week of hydrogen production at full nominal capacity. Specifically, the storage capacity has been calculated considering a nominal production flow of 0.1 kg/s per electrolyser module, with fifteen modules operating simultaneously, resulting in a weekly storage requirement of approximately 100 tonnes:

$$\text{Storage capacity } [\text{kg}] = 0.1 \left[\frac{kg}{s}\right] \times 0.1\,[-] \times 15\,[\#] \times 3600 \left[\frac{s}{h}\right] \times 24 \left[\frac{h}{day}\right] \times 0.1\,[-] \times 7\,[day] \approx 100\text{ton} \quad (11)$$

Based on this storage capacity, the total CAPEX for the hydrogen buffer is estimated at approximately 49 million EUR (TANDEM Project Documentation). Similar to the assumptions made for the HTSE units, the hydrogen buffer is assumed to have a technical lifetime of 20 years, implying that it will need to be replaced twice during the 60-year project horizon, and the same cost reduction assumptions adopted for HTSE technology are applied.

5.2.4 Capex for Hydrogen Compressor

CAPEX have been still estimated based on data from the TANDEM project. The specific investment cost is reported as 1,000,000 EUR per MW of electrical power input, and the required compressor capacity has been determined using TANDEM's reference, which indicates that compressing a hydrogen mass flow rate of 8,257 kg/h requires 17.82 MW of electrical power. Adapting this ratio to mass flow rates results in a required compressor capacity of approximately 1.5 MW, leading to a total CAPEX of around 1,500,000 EUR. The compressor is assumed by TANDEM to have a lifetime of 10 years, and the same cost reduction dynamics assumed for HTSE units have been applied.

5.2.5 Capex for Gas Turbine

The CAPEX for the gas turbine LM6000 have been estimated based on data reported in the study by Riboldi et al. (2020), which analyzed the environmental and economic impacts of offshore oil platform electrification. According to this study, the investment cost for one LM6000 gas turbine is approximately 23.2 million EUR. Although the system design requires a gas turbine capacity of 47 MW, to adopt a conservative approach, cost figures were retained corresponding to the 57 MW LM6000 unit as reported in the article, given that the model and technology are identical to those implemented in the selected design. The operational lifetime of the LM6000 gas turbine is assumed to be approximately 20 years.

5.2.6 Capex for Methanation Reactor

The CAPEX for the methanation reactor have been estimated at approximately 8.19 million EUR. This value has been derived based on the work of Götz et al. (2016), who reported that the investment cost of a methanation unit is roughly one-thirtieth of the cost of electrolysers. Accordingly, the same ratio was applied to specific electrolyser CAPEX figures to estimate the cost of the methanation subsystem in our project. The methanation reactor is assumed to have a technical lifetime of 20 years, and similarly as the HTSEs the methanation reactor is expected to benefit from cost reductions over time as the technology matures.

5.3 OPEX

The estimation of OPEX has been carried out primarily based on data and assumptions provided in the TANDEM project documentation, with few exceptions of other sources. This analysis includes costs related to operation, maintenance, consumables, and other recurring expenses over the plant's lifetime.

5.3.1 Opex for E-SMRs

The OPEX for the E-SMRs are divided into variable costs, which amount to 24.4 EUR2023 per MWh of electrical output, and fuel costs are reported as 7.4 EUR2023 per MWh (TANDEM Project Documentation, 2023). Therefore, the total operating cost for the e-SMR system is calculated as 31.8 EUR2023/MWhe. Considering a combined power of 354 MW for the two E-SMR units, operating for 8,760 hours per year at a load factor of 0.98, the annual OPEX amounts to approximately 96,640,811 EUR:

$$\text{Annual OPEX [EUR/year]} = 354 \times 8{,}760 \times 0.98 \times 31.8 \approx 96{,}640{,}811 \tag{12}$$

Moreover, as highlighted in Riboldi et al. (2024), OPEX are expected to decrease over time as the technology transitions from First-of-a-Kind (FOAK) to Nth-of-a-Kind (NOAK) status. After the initial 15 years of operation, costs are projected to decrease by approximately 30% due to increased technological maturity and operational experience. During the subsequent 15 years, a further reduction of 10% is expected, reflecting continued but slower learning curve effects. Beyond the first 30 years, OPEX levels are expected to stabilize, approaching their asymptotic lower limit. Unlike CAPEX reductions, these OPEX decreases have been modeled as gradual annual reductions.

5.3.2 Opex for HTSEs

The OPEX for the HTSE have been set to 2.5% of the corresponding CAPEX (TANDEM Project Documentation, 2023), corresponding to an annual OPEX of approximately 6.1425 million EUR. The OPEX for HTSE units is expected to annually decrease over time with the same cost reduction dynamics assumed for CAPEX, namely a 30% reduction after the first 20 years and an additional 15% reduction in the following 20 years.

5.3.3 Opex for Hydrogen Storage Buffer and Compressor

Annual OPEX for the Buffer are considered equal to 1.5% of the CAPEX, while those for the hydrogen compressor are equal to 1% of the CAPEX (TANDEM Project Documentation, 2023), resulting in respectively 735,000 and 15,000EUR per year. As with the electrolysers, the OPEX for hydrogen storage is expected to decrease over time, following the same cost reduction trends applied to the CAPEX of hydrogen technologies.

5.3.4 Opex for Gas Turbine

OPEX have been estimated using data from Gas Turbine World (2024), which reports fixed O&M costs of 16.30 USD/kW, resulting in about 13,930 EUR/MW annually, after conversion from USD and applying an exchange rate of 1.17 USD/EUR. Therefore, the total annual OPEX for the LM6000 gas turbine would be approximately 654,710 EUR. Since the project already secures its fuel supply from other subsystems, we have conservatively assumed that fuel-related OPEX can be reduced by 50%, leading to the final estimate of 327,355 EUR.

5.3.5 Opex for Methanation Reactor

The OPEX of a Methanation Reactor are significantly influenced by the costs of its input streams, particularly hydrogen and carbon dioxide (CO_2). While hydrogen is produced on-site, the required CO_2 must be sourced externally. It has been assumed that the acquired CO_2 is captured from industrial processes, where capture costs largely depend on the concentration of CO_2 in the flue gas stream, since higher CO_2 concentrations generally reduce capture costs.

Two primary references have been considered for estimating CO_2 capture costs, which were used as proxies for C02 price. The first, the Global CCS Institute report (2017), indicates costs ranging from 30 to

70 $\frac{EUR}{ton-CO_2}$ depending on the industrial source, from coal-based to natural gas processes. The second study by Smith et al. (2024) reports an average cost of approximately 60 $\frac{EUR}{ton-CO_2}$, although without specifying the industrial source. Based on these data, a mean CO_2 capture cost of 55 $\frac{EUR}{ton-CO_2}$ has been assumed as a balanced compromise between the reported values.

Furthermore, future projections for CO_2 capture costs have been informed by findings from Gruber et al. (2022), who suggest that ongoing research and technological improvements are expected to reduce these costs of approximately 73.5% (from 95 USD/ton to 25 in the long term), leading the initial 55 $\frac{EUR}{ton-CO_2}$ to an estimated cost of roughly 14.5 $\frac{EUR}{ton-CO_2}$ by around 2085. This cost reduction has been implemented as a gradual decrease over time.

5.4 Revenues

The revenue streams originate from multiple products and market segments. Specifically, revenues are generated through the sale of electricity to the local energy community (Comunità Energetica Rinnovabile, CER) and, when production exceeds local demand, through sales to the wholesale electricity market. Additionally, revenues are derived from hydrogen sales targeted primarily at the transport sector, as well as from synthetic gas (syngas) sales intended to meet the residual fossil fuel demand within the region and, in case of surplus production, to external markets beyond the project's geographical scope.

5.4.1 Electricity Revenues

For electricity, revenues have been computed as the sum of earnings from sales to the CER and revenues from surplus electricity dispatched to the wholesale market.

Revenues from sales to the CER have been determined by considering the total annual electricity output delivered to the community, as calculated by the Modelica simulation outputs. The electricity price applied for these calculations has been derived from historical data of the zonal day-ahead electricity price (PUN zonale) for the Northern Italian market area, adjusted by a discount factor to promote public acceptance of the infrastructure. Specifically, the reference electricity price has been established by analyzing the monthly PUN zonale data recorded from 2022 to the present(GSE, 2023). An initial estimation of the zonal price for the remaining months of 2025 was obtained by extrapolating the trend observed in 2023 and 2024 through linear interpolation. Subsequently, a weighted average monthly price over the past three years was calculated, assigning higher weights to more recent data to reflect current market dynamics. Finally, the average annual price was determined by computing a weighted mean of these monthly values, where the weighting was based on the quantity of electricity sold to the community in each month, since this parameter is strongly influenced by seasonal variations in the community's photovoltaic self-generation capacity. To enhance public acceptance and encourage regional engagement, a 10% discount was applied to the final price offered to the CER as a form of implicit subsidy.

For electricity sold to the wholesale market, the same methodology was adopted for price estimation, following identical steps in the analysis of historical and projected PUN zonale data, since it is assumed that the energy surplus will be distributed in Lombardy. However, in this case, the 10% discount was not applied, as market sales are not subject to local incentives or community agreements.

Furthermore, long-term electricity price evolution has been informed by the work of Gabrielli et al. (2022), where in a scenario modeled for the UK electricity market in 2035, the study predicted a moderate growth. Based on these findings and the graphical results presented in the paper, we have assumed a cumulative electricity price increase of approximately 10

5.4.2 Hydrogen Revenues

Hydrogen is commonly categorized based on its production pathway, which determines both its carbon footprint and market value. Among the different classifications, green hydrogen is produced through water electrolysis powered exclusively by renewable energy sources, while hydrogen produced via nuclear-powered electrolysis (referred to as pink or purple hydrogen) utilizes low-carbon electricity from nuclear reactors.

Due to a lack of robust market data on nuclear-derived hydrogen, the selected price has been approximated using the market value of green hydrogen, which was identified at approximately 240 EUR per MWh (EEX, 2025). This corresponds to a price of slightly over 8.5 EUR per kg of hydrogen. Consistent with the assumptions made for the declining production costs of hydrogen over time, the same approach has been applied to the projected sales price. The only difference with respect to the previous assumptions is that, for the final 20 years of the analysis, hydrogen is considered a sufficiently mature commodity whose price dynamics are expected to align with those of electricity, thereby following the same trend as electricity prices in the long term.

5.4.3 Synthethic Gas Revenues

Revenues from syngas sales have been estimated using data from the TANDEM project, which reports a market price of approximately 120 EUR per MWh for synthetic methane or syngas. Given that syngas production relies on hydrogen and captured CO_2 as input streams, which are projected to have a similar evolution in term of price reduction over time, its cost structure and market price are inherently linked to the dynamics of these two commodities. Consequently, the price of syngas is expected to evolve similarly to that of hydrogen. As with hydrogen, it has been assumed that in the final 20 years of the analysis, syngas will be a mature commodity whose price trends will align more closely with those of electricity and its associated energy markets.

5.5 Economic Results: Base Solution

For the economic evaluation of the base case scenario, in which the hybrid energy system is implemented from the beginning, several financial performance indicators have been calculated, including the Net Present Value (NPV), the Levelized Cost of Electricity (LCOE), and the Internal Rate of Return (IRR). To perform these calculations, the economic model incorporates all cost and revenue data discussed in the previous sections, enabling the reconstruction of a comprehensive income statement and cash flow statement over the entire project horizon.

A discount rate of 5

Consistent with prior assumptions in the economic model, the first project year, 2025, is entirely dedicated to capital investment expenditures, which for this base case scenario amount to a total of 2,469,290,000 EUR. As anticipated, the NPV over the considered project horizon results significantly negative, with a calculated value of approximately -982 million EUR and an Internal Rate of Return (IRR) of about 2.54

5.5.1 Sensitivity Analysis

To enhance the robustness of the economic assessment, a sensitivity analysis was performed to evaluate how variations in CAPEX and the discount rate affect the Net Present Value (NPV) and Levelized Cost of Energy (LCOE) of the fully hybrid system.

Results presented in the Appendix show that achieving NPV break-even would require CAPEX to drop to around 1,487 million EUR, with more attractive scenarios falling in the 800–1,200 million EUR range—levels currently unrealistic for a facility of this scale and technology maturity. Similarly, the LCOE would need to decrease to approximately 115 EUR/MWh to reach economic viability under these conditions.

Regarding the discount rate, values between 0.5

5.6 Economic Results: Alternative Solution

To further explore viable pathways, an alternative "middle-of-the-road" scenario was analyzed. Initially, a full-electric configuration was simulated in which only electricity is produced and sold, and solely the capital and operational costs of the E-SMR reactors are considered. This scenario yielded highly positive results, with an NPV of approximately 2.5 billion EUR and an LCOE of 61.5 EUR/MWh.

However, as previously discussed in the case study section, the EU project aims to finance hybrid system solutions. Therefore, a purely full-electric approach, despite its favorable economics, is not feasible from a policy and funding perspective. This underscores the need for a balanced intermediate solution that satisfies both technical and financial constraints.

The chosen "middle-of-the-road" approach foresees operating exclusively in full-electric mode during the first 20 years, with only the CAPEX and OPEX of the E-SMRs accounted for in 2025. From 2046 onwards, the remaining parts of the hybrid system are integrated. This scenario results in a profitable outcome, delivering an NPV of approximately 1,160 million EUR with a payback time of 14 years, an IRR of 9.03

5.7 Sensitivity Analysis: Alternative Solution

The same sensitivity analysis methodology used for the base case was applied to the middle-of-the-road scenario, providing clear evidence of the robustness of this more balanced approach.

Analyzing the NPV as a function of CAPEX shows that the solution is resilient, given the breakeven CAPEX of approximately 3,302 million EUR, which is a velue well above the actual investment made. Similar conclusions emerge when examining the LCOE relative to CAPEX: the breakeven LCOE is around 100 EUR/MWh, while the observed LCOE from the analysis is significantly lower, at 78.34 EUR/MWh.

Consistent results are observed when assessing the NPV and LCOE as functions of the discount rate, the break even value of which is equal to 9,3

6 ENVIRONMENTAL IMPACT

The environmental impact of the plant is assessed based on the following indicators: GHG emissions intensity, water usage, and land use.

The assessment is conducted on the optimized operational scenario (previously named "Middle of the Road") in which, for the first 20 years, both reactors are dedicated exclusively to electricity production and sale.

6.1 GHG emissions intensity assessment

6.1.1 Assumptions and Methodology

GHG emissions intensity are evaluated considering:

- The impact on the regional energy mix (Lombardy), which is expected to be minor.
- The impact on the local mix within the Seven Lakes Region, where the plant has a significantly more noticeable influence.
- Emissions related to residual fossil fuel demand for building heating.
- Emissions from the transportation sector, expected to decrease due to the hydrogen supplied by the plant directly to this sector.

6.1.2 Data Sources

GHG emissions intensities The data for the current (2023) Lombardy energy mix were sourced from Terna, including imported energy shares. GHG emissions intensity per energy source are based on mean values [13]. Specifically, solar photovoltaic (PV) was considered for the "Renewable" share, as it is the most prevalent renewable source and the one of major interest for this study.

The values used are:

Energy Source	GHG Intensity (gCO_2eq/kWh)
Coal	996
Natural Gas	550
Oil	780
Nuclear	12
Renewables	56

Table 2: *GHG emissions intensities by energy source.*

For nuclear power, other sources support values around 12 gCO_2eq/kWh [14] and [15].

Equivalent GHG emissions intensity for the other plant products are:

Product	GHG Intensity (gCO_2eq/kWh)
Hydrogen	19.39
Syngas	72.76

Table 3: *Equivalent GHG emissions intensity for plant products*

Hydrogen carbon intensity assumes electricity for HTSE is fully supplied by the two SMRs and uses its LHV for evaluating the energy equivalent of hydrogen produced. Syngas is produced with that same hydrogen, produced with nuclear-provided electricity. The CO_2 is not considered in the GHG intensity of syngas, as it was explained in the economic section.

Energy consumption Data The baseline energy data prior to year 0 are derived from current statistics, with total electricity consumption and imports for Lombardy taken from Terna. Values are computed every 15 years to reflect assumed changes in building energy classifications and related energy demands. An exception is made for year 20 to account for the plant's secondary section's commissioning.

A key assumption is that the electricity produced by the proposed plant displaces imported electricity—an economically rational choice. The renewable share varies across scenarios, reflecting the increasing photovoltaic (PV) penetration over time.

6.1.3 Calculations

Regional Energy Mix The regional impact is assessed in terms of how the plant alters Lombardy's energy mix, which is expected to be minimal due to its limited size (maximum output of 3 TWh/year vs. 65 TWh/year of yearly regional consumption [1]). Nevertheless, the impact is more significant in the first 20 years when the plant contributes only electricity.

The regional mix GHG intensity is calculated as:

$$\left(\frac{gCO2eq}{kWh}\right)_{mix} = \sum_i \left(\left(\frac{gCO2eq}{kWh}\right)_i \cdot \chi_i\right) \tag{13}$$

where i indexes each energy source and each energy source's share χ_i is taken into account.

Local Energy Mix Using a methodology analogous to the one applied at the regional level, the electricity consumption of the Seven Lakes Region is estimated based on monthly energy balances and photovoltaic production data previously gathered. The local energy mix is reconstructed using three main components: Nuclear, Renewables, and Fossil. The evolution of this mix over time, and its impact on GHG emissions intensity, is evaluated using the previous approach.

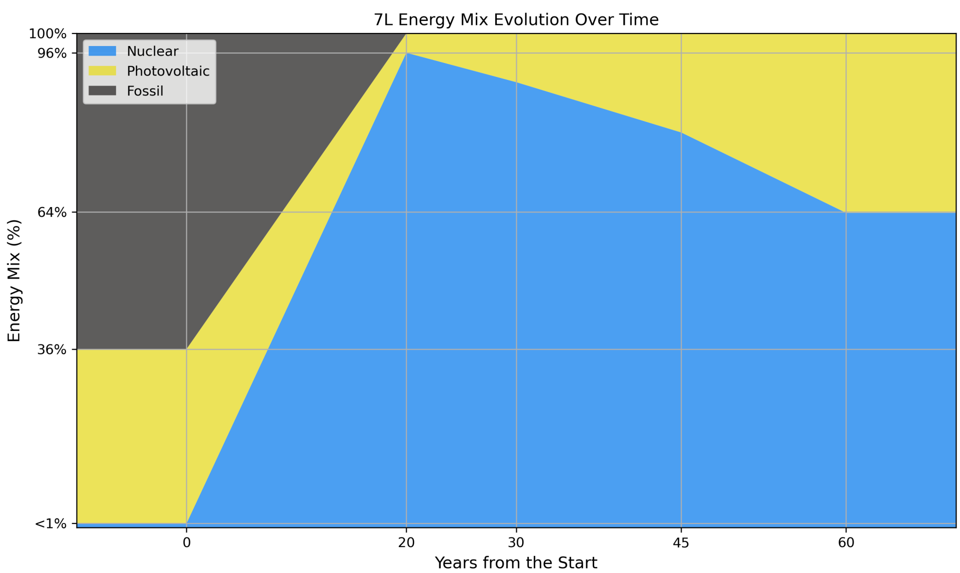


Figure 15: *Local energy mix evolution over time.*

Residual Fossil Fuel Consumption The GHG emissions intensity associated with the residual fossil fuel demand for residential heating are estimated using the processed data, alongside the amount of syngas exported to the market.

The gCO_2eq/kWh intensity for this sector is calculated by weighting:

Fuel Type	**GHG Intensity (gCO$_2$eq/kWh)**
Syngas	72.76
Natural Gas	550

Table 4: *GHG intensities for syngas and natural gas in the residential heating sector.*

This impact is zero in the first 20 years of operation, as the plant only provides electricity. From year 21, the introduction of syngas production leads to a growing influence on this sector.

Fossil fuel demand is projected to decrease over time due to improved building energy performance and increased photovoltaic (PV) penetration, while syngas production remains constant — resulting in a gradually increasing contribution to GHG mitigation.

Transportation Sector Transport energy consumption data for the region were estimated by scaling national data [16] according to the population share of the Seven Lakes Region with respect to the national total [4].

GHG intensities for diesel and gasoline (currently dominant in ground transport) were sourced from [17]:

Fuel Type	**GHG Intensity (gCO$_2$eq/kWh)**
Diesel	266.5
Gasoline	263

Table 5: *GHG intensities for diesel and gasoline.*

The impact of the hydrogen produced by the plant is assessed by assuming that it is sold directly to the transportation sector. The resulting GHG intensity is calculated based on the share of energy demand met by:

Energy Carrier	**GHG Intensity (gCO$_2$eq/kWh)**
Hydrogen	19.39
Fossil fuels (avg. diesel/gasoline)	264.75

Table 6: *GHG intensities for hydrogen and fossil fuels in the transportation sector.*

Beyond the GHG reduction, it is worth highlighting that hydrogen substitution would also positively affect local air quality due to the elimination of non-GHG emissions typically associated with diesel and gasoline combustion. However, this benefit is not quantified within this study.

6.1.4 Results

All results are summarized in the following plots, which illustrate the evolution of GHG emission intensities over time, along with changes in electricity and fossil fuel demand.

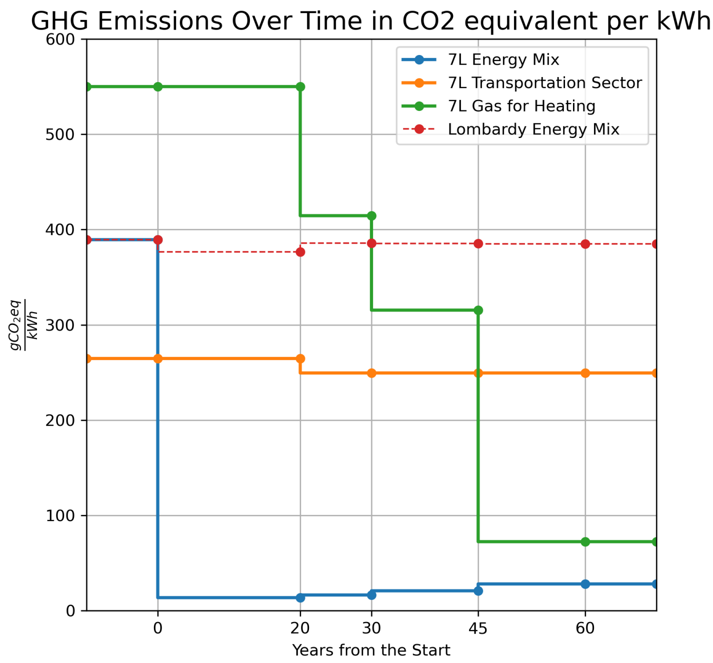


Figure 16: *GHG emissions intensity over time.*

Figure 17: *Local Net Electricity and Fossil Demand evolution over time.*

The plots highlight several significant trends, including:

- ...
- ...
- ...

6.2 Land Use

One of the key advantages of the proposed project is its minimal land use impact. The compact scale of the facility, combined with its placement in an area already approved for nuclear activities, ensures efficient use of space. The site is located within the Joint Research Centre (JRC) complex, which already hosts research reactors (ESSOR and Ispra-1) and associated nuclear infrastructure. Notably, a containment building is available on-site and is deemed suitable to house at least one of the two planned Small Modular Reactors (SMRs), greatly reducing the need for new construction and environmental disturbance.

7 LIMITATIONS AND POSSIBLE IMPROVEMENTS

7.1 Data Collection

If the data for each class were obtained from a set of real buildings in the region instead of just one, it would yield better results. Ideally, a statistically significant set of buildings and variable utilization profile could be used. Moreover, it would be beneficial to weigh the simulation data on the real energy requirements of those buildings. This is a much more complex task since it would require extensive data acquisition throughout the year.

From the simulation, humidification and dehumidification electricity needs are neglected since these plants are rarely installed today. A better evaluation would consider the possible future share of air treatment plants in the residential sector. For the purpose of this study, the share of installed air conditioning systems for cooling was assumed constant throughout the years, but it might increase in the future.

7.1.1 Renewable Energy Sources

Solar-thermal systems were neglected since they are considered not very impactful and complex to evaluate.

7.1.2 Conversion to Primary Energy

Time-dependent efficiency for heat pumps, taking into account the ambient temperature could improve the model.

7.1.3 Other Electricity Demand

This is the most complex contribution to evaluate, especially considering that it is not easy to forecast how electricity consumption will evolve in 60 years. Collecting a statistically significant number of data from the region regarding the type of appliances installed and their consumption would improve the simulation. For the purpose of this study the appliances demand was assumed constant throughout the years.

7.2 Plant Modelling and Simulation

The Modelica model could be utilized to analyze the dynamic response of the plant to transient conditions, thereby evaluating the system's adequacy and frequency control capabilities. This analysis could focus on critical periods, such as specific days of the year characterized by significant demand oscillations.

Additionally, buffers with limited capacity could be modeled to further understand their impact on system dynamics. The hydrogen plant could be simulated to partially shut down some HTSE units, allowing for better adaptation to demand variations.

Incorporating chemical kinetics into the methanation model could enhance the simulation of the plant's dynamic response to transient conditions. Furthermore, the optimization of hydrogen tapped for market consumption could be based on a detailed analysis of the transportation sector.

Lastly, the sizing of the entire plant should be optimized based on a comprehensive economic assessment.

7.3 Economic Assessment

7.3.1 Impact of interest rates on borrowed capital

7.3.2 Impact of water cost on HTSE OPEX

7.4 Environmental Impact Assessment

7.4.1 Water Usage

The impact on water resources, primarily required for reactor and Balance of Plant (BoP) operation as well as for the High Temperature Steam Electrolyzer (HTSE) units, is deemed negligible. This is due to the plant's location near Lake Maggiore—one of Italy's largest lakes—which ensures a stable water supply without significant disruption to the basin's hydrological balance or ecosystem.

7.4.2 Resource Depletion

8 CONCLUSIONS

The project demonstrates that hybrid energy systems can significantly reduce carbon emissions in the studied area by fully eliminating fossil fuel use for electricity generation. Moreover, it lays the groundwork for further decarbonization of the transportation and heating sector.

The plant model confirms the technical feasibility of the proposed system, providing a detailed, hour-by-hour account of energy and material flows both within the plant boundaries and between the plant and the surrounding community.

The plant layout was, at first, roughly designed primarily around the operational requirements of the gas turbine. As a result, the number of High-Temperature Steam Electrolyzers (HTSEs) and Small Modular Reactors (SMRs) was determined based on this component constraints. The plant was accordingly oversized, in order to ensure sufficient syngas production to let the gas turbine operate at full capacity.

The ambitious decarbonization goals led to a substantial portion of the nuclear-generated electricity being allocated to low-profit decarbonization processes — such as hydrogen and methane production — rather than being sold directly to the grid.

A key finding of the economic analysis is that the high initial capital costs of the full hybrid system are unlikely to be recovered if all components are built simultaneously. However, a phased implementation strategy can mitigate these costs and ultimately generate positive returns.

To manage the financial burden and benefit from advancements in relevant technologies, the installation of the HTSE-Methanator-Gas Turbine complex is postponed until the 20th year of operation. In the first phase, the electricity from the SMRs is sold directly to the grid, generating more immediate and profitable returns to help offset nuclear capital expenditures.

This phased strategy mirrors the real-world approach of deploying SMRs incrementally to finance subsequent units. Moreover, this strategic choice demonstrates resilience with respect to the main financial parameters possibly affecting its economic viability.

When combined with the project's strong decarbonization potential and the elimination of foreign energy dependence, these results make the proposal a compelling opportunity for the Seven Lakes region and a strategic use of the JRC's nuclear infrastructure.

APPENDIX

```
model MethanationReactor
  import      Modelica.Units.SI;
  import Modelica.Blocks.Interfaces.RealInput;
  import Modelica.Blocks.Interfaces.RealOutput;

  // Parameters
  parameter SI.Volume V = 0.01 "Reactor volume [m^3]";
  parameter SI.Temperature T = 573.15 "Reactor temperature [K]";
  parameter SI.Pressure p = 1e5 "Operating pressure [Pa]";
  parameter Real A = 1e10 "Pre-exponential factor [mol/(m^3.s)]";
  parameter Real Ea = 100000 "Activation energy [J/mol]";
  parameter Real conversionEfficiency = 0.80 "CO2+H2 conversion efficiency";

  // Inputs: molar flow rates [mol/s]
  RealInput n_H2_in "Inlet molar flow rate of H2"
  RealInput n_CO2_in "Inlet molar flow rate of CO2"

  // Outputs
  RealOutput PCI "Lower calorific value [MJ/kg]"
  RealOutput m_flow_rate "Total gas mass flow rate [kg/s] (CO2+CH4+H2)"

  // Constants
  constant Real R = 8.314 "Gas constant [J/mol.K]";
  constant Real M_CH4 = 16.0 "CH4 molar mass [g/mol]";
  constant Real M_CO2 = 44.0 "CO2 molar mass [g/mol]";
  constant Real M_H2  = 2.0  "H2 molar mass [g/mol]";
  constant Real LHV_CH4 = 50.0e6 "CH4 PCI [J/kg]";
  constant Real LHV_H2  = 120.0e6 "H2 PCI [J/kg]";
  constant Real UHV_CH4 = 55.4e6 "CH4 PCS [J/kg]";
  constant Real UHV_H2 = 141.8e6 "H2 PCS [J/kg]";

  // Internal variables
  Real n_CH4_out, n_H2_out, n_CO2_out;
  Real m_CH4_out, m_H2_out, m_CO2_out;

  // Temporary variable
  Real CH4_formed;

  RealOutput PCS "Upper calorific value [MJ/kg]"
algorithm
  // Conversion based on limiting reagent
  CH4_formed := min(n_CO2_in, n_H2_in / 4) * conversionEfficiency;

  // Molar flows after conversion
  n_CH4_out := CH4_formed;
  n_H2_out  := max(n_H2_in - 4 * CH4_formed, 0);
  n_CO2_out := max(n_CO2_in - 1 * CH4_formed, 0);

  // Mass flows [kg/s]
  m_CH4_out := n_CH4_out * M_CH4 / 1000;
  m_H2_out  := n_H2_out  * M_H2  / 1000;
  m_CO2_out := n_CO2_out * M_CO2 / 1000;

  m_flow_rate := max(m_CH4_out + m_H2_out + m_CO2_out, 1e-6); // stabilized

  // Calorific value [MJ/kg]
  PCI := max((m_CH4_out*LHV_CH4 + m_H2_out*LHV_H2) / m_flow_rate, 0);
  PCS := max((m_CH4_out*UHV_CH4 + m_H2_out*UHV_H2) / m_flow_rate, 0);

end MethanationReactor;
```

Listing 1: *Methanation Reactor*

```
  model MethanationControlInput
  import      Modelica.Units.SI;
  import Modelica.Blocks.Interfaces.RealInput;
  import Modelica.Blocks.Interfaces.RealOutput;

  // Parameters
  parameter Real stoichRatio = 4 "H2:CO2 molar ratio (default = 4 for stoichiometric)";
  parameter SI.Time T = 0 "Response time for control ramp [s]";

  // Constants
  constant Real M_H2 = 2.0 "H2 molar mass [g/mol]";

  // Inputs
  RealInput m_H2_in "H2 mass flow rate [kg/s]"

  // Outputs
  RealOutput n_H2_out "H2 molar flow rate [mol/s]"
  RealOutput n_CO2_out "CO2 molar flow rate [mol/s]"

  // Internal variables (target values)
  Real n_H2_target;
  Real n_CO2_target;

  // State variables for first-order delay
  Real n_H2_delayed(start=0);
  Real n_CO2_delayed(start=0);

equation
  // Convert to molar flow
  n_H2_target = m_H2_in * 1000 / M_H2;
  n_CO2_target = n_H2_target / stoichRatio;

  if T > 0 then
    der(n_H2_delayed) = (n_H2_target - n_H2_delayed) / T;
    der(n_CO2_delayed) = (n_CO2_target - n_CO2_delayed) / T;
  else
    n_H2_delayed = n_H2_target;
    n_CO2_delayed = n_CO2_target;
  end if;

  // Outputs
  n_H2_out = n_H2_delayed;
  n_CO2_out = n_CO2_delayed;
end MethanationControlInput;
\
```

Listing 2: *Methanation Control Input*

```
  model H2_Buffer
  Modelica.Blocks.Interfaces.RealInput mass_flow_in
  Modelica.Blocks.Interfaces.RealOutput mass_flow_out
  parameter Real stock = 0;

  Real Content(start=0) "Amount of mass stored in the buffer";

equation
    // Mass balance differential equation
  der(Content) = mass_flow_in - mass_flow_out;

  // Ensure mass_flow_out does not exceed the available content
  mass_flow_out = if Content > 0 then min((1-stock)*mass_flow_in, Content) else 0;

end H2_Buffer;
```

Listing 3: *H2 Buffer*

```
  model CH4_Buffer
  Modelica.Blocks.Interfaces.RealInput mass_flow_in
  Modelica.Blocks.Interfaces.RealInput mass_flow_out
  Modelica.Blocks.Interfaces.RealInput mass_flow_to_grid

  Real Content(start=0) "Amount of mass stored in the buffer";

equation
  // Mass balance differential equation
  der(Content) = mass_flow_in - mass_flow_out - mass_flow_to_grid;

end CH4_Buffer;
```

Listing 4: *CH4 Buffer*

```
  model Control_Room
  import Modelica.Blocks.Types.Extrapolation;
  import Modelica.Blocks.Sources.CombiTimeTable;

  parameter Real n_HTSE = 8;
  parameter Real manualTable[:, :] = fill(0.0, 2, 5);
  parameter Real n_Reactors = 1;

  // Inputs
  Modelica.Blocks.Interfaces.RealInput P_nuc(start=100e6);
  Modelica.Blocks.Interfaces.RealInput P_turbine(start = 10e6);
  Modelica.Blocks.Interfaces.RealInput P_HTSE( start =1e6);

  // Outputs
  Modelica.Blocks.Interfaces.RealOutput setpoint_H2;
  Modelica.Blocks.Interfaces.RealOutput mflow_tapSteam;
  Modelica.Blocks.Interfaces.RealOutput turbine_load;
  Modelica.Blocks.Interfaces.RealOutput fossil_fuel_demand;

  // Variables
  Real Energy_Excess(start=0);
  Real tap_Setpoint(start=0);
  Real ramped_HTSE;
  Real ramped_turbogas;
  Real total_fossil_fuel_demand;
  Real Demand;

  // Load CSV data
  CombiTimeTable loadProfile(
  tableOnFile = false,
  table = manualTable,
  columns = {2,3,4,5},
  timeScale = 3600,
  tableName = "",
  extrapolation = Modelica.Blocks.Types.Extrapolation.HoldLastPoint)

equation
  // Output assignments from table
  ramped_HTSE              = loadProfile.y[1];
  ramped_turbogas          = loadProfile.y[2];
  Demand                   = loadProfile.y[3];
  total_fossil_fuel_demand = loadProfile.y[4];


  // GUI-visible outputs
  setpoint_H2          = ramped_HTSE / 10;
  turbine_load         = ramped_turbogas;
  fossil_fuel_demand   = total_fossil_fuel_demand;
  tap_Setpoint         = 2 * setpoint_H2;
  mflow_tapSteam       = max(0.001, tap_Setpoint);

  // Energy balance
  Energy_Excess        = n_Reactors * P_nuc + P_turbine - P_HTSE * n_HTSE - Demand*1e6;

end Control_Room;
```

Listing 5: *Control Room*

REFERENCES

[1] Terna SpA, *Dati Statistici sui Consumi di Energia Elettrica per Settore*, https://dati.terna.it/fabbisogno/dati-statistici, Accessed: 2025-06-26, 2025.

[2] Regione Lombardia, *Database CENED+2 - Certificazione ENergetica degli EDifici*, https://www.dati.lombardia.it/Energia/Database-CENED-2-Certificazione-ENergetica-degli-E/bbky-sde5, Accessed: 2025-03-25, 2025.

[3] Gestore dei Servizi Energetici (GSE), *Mappa Interattiva delle Cabine Primarie*, https://www.gse.it/servizi-per-te/autoconsumo/mappa-interattiva-delle-cabine-primarie, Accessed: 2025-04-14, 2025.

[4] Istituto Nazionale di Statistica (ISTAT), *Censimento permanente della popolazione - Anno 2023 - Lombardia*, https://www.istat.it/wp-content/uploads/2025/04/Censimento-permanente-popolazione_Anno-2023_Lombardia.pdf, Accessed: 2025-06-26, 2025.

[5] E. Fuentes, L. Arce, and J. Salom, "A review of domestic hot water consumption profiles for application in systems and buildings energy performance analysis," *Renewable and Sustainable Energy Reviews*, vol. 81, pp. 1530–1547, 2017. DOI: 10.1016/j.rser.2017.05.229.

[6] OpenStreetMap Contributors, *Nominatim Search Tool*, https://nominatim.openstreetmap.org/ui/search.html, Accessed: 2025-04-16, 2025.

[7] Joint Research Centre, *Photovoltaic Geographical Information System (PVGIS)*, https://re.jrc.ec.europa.eu/pvg_tools/en/, Accessed: 2025-04-16, 2025.

[8] Acquirente Unico S.p.A., *Electricity residential demand*, https://www.consumienergia.it/portaleConsumi/it/energia-elettrica-utenze-domestiche.page, Accessed: 2025-04-14, 2025.

[9] G. Simonini et al., "Integrating Small Modular Reactors into hybrid energy systems: the tandem Modelica library," in *International Conference on Small Modular Reactors and their Applications*, IAEA, Vienne, Austria, Oct. 2024. [Online]. Available: https://hal.science/hal-04645663.

[10] Politecnico di Milano, *Modelica models description for the "TANDEM" library*, https://tandemproject.eu/wp-content/uploads/2024/07/D2_3_Modelica_models_description_for_the_tandem_library_V1-1.pdf, Accessed: 2025-04-16, 2024.

[11] M. Götz et al., "Renewable power-to-gas: A technological and economic review," *Renewable and Sustainable Energy Reviews*, vol. 53, pp. 1472–1482, 2016, ISSN: 1364-0321. DOI: https://doi.org/10.1016/j.rser.2015.09.042. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0960148115301610.

[12] Dassault Systèmes, *Dymola, Dynamic Modeling Laboratory*, https://www.3ds.com/products/catia/dymola, Accessed: 2025-04-16, 2025.

[13] D. Weisser, "A guide to life cycle greenhouse gas (ghg) emissions from electric supply technologies," *Energy*, vol. 32, no. 9, pp. 1543–1559, 2007. DOI: 10.1016/j.energy.2007.01.008.

[14] V. et al., "Life cycle analysis of a network of small modular reactors," in *IOP Conference Series: Earth and Environmental Science*, International Conference on Sustainable Energy and Green Technology 2023, vol. 1372, IOP Publishing, 2024, p. 012 059. DOI: 10.1088/1755-1315/1372/1/012059.

[15] L. Wang et al., "A comparative life-cycle assessment of hydro-, nuclear and wind power: A china study," *Applied Energy*, vol. 249, pp. 37–45, 2019. DOI: 10.1016/j.apenergy.2019.04.099. [Online]. Available: https://doi.org/10.1016/j.apenergy.2019.04.099.

[16] Eurostat, *Energy balance flow diagram (Sankey) – Italy, 2023*, https://ec.europa.eu/eurostat/cache/sankey/energy/sankey.html?geos=IT&year=2023&unit=GWh&fuels=TOTAL&highlight=_&nodeDisagg=1111111111111&flowDisagg=true&language=EN, Accessed: 2025-07-01, 2023.

[17] V. Quaschning, *Specific carbon dioxide emissions of various fuels*, https://www.volker-quaschning.de/datserv/CO2-spez/index_e.php, Creative Commons license, accessed 1 July 2025, 2025.